

SAPIENZA
UNIVERSITÀ DI ROMA

Search for New Physics in anomalous jets using jet substructures with the ATLAS detector and LHC Run 3 data

Facoltà di Scienze Matematiche, Fisiche e Naturali
Master's Degree in Physics

Valerio Tinari
ID number 1998628

Advisor
Prof. Stefano Giagu

Co-Advisors
Dr. Valerio Ippolito
Dr. Giuliano Gustavino

Academic Year 2025/2026

Thesis defended on 12/07
in front of a Board of Examiners composed by:

examiner (chairman)

Search for New Physics in anomalous jets using jet substructures with the ATLAS detector and LHC Run 3 data

Sapienza University of Rome

© 2026 Valerio Tinari. All rights reserved

This thesis has been typeset by LaTeX and the Sapthesis class.

Author's email: tinari.1998628@studenti.uniroma1.it

dedica

Contents

Introduction **1**

1 (Beyond) The Standard Model **4**
1.1 An overview of the Standard Model 4
1.1.1 Quantum Electrodynamics 6
1.1.2 Quantum Chromodynamics 7
1.1.3 Electroweak model 8
1.1.4 The Brout-Englert-Higgs mechanism 9
1.2 Hints of Physics Beyond the Standard Model 10
1.2.1 New Physics at colliders 11
1.2.2 Heavy Vector Triplets 12
1.2.3 Dark Jets 13

2 The LHC and the ATLAS experiments **14**
2.1 The LHC 14
2.1.1 CERN's accelerator system 15
2.1.2 The LHC parameters 16
2.1.3 LHC Run Schedule 17
2.2 *pp* collisions at the LHC 18
2.2.1 Pile-Up and hadronisation 19
2.2.2 Electromagnetic Showers 20
2.2.3 Hadronic Showers 21
2.3 The ATLAS detector 21
2.3.1 Coordinate system 21
2.3.2 Magnetic system 24
2.3.3 Particle tracking 25
2.3.4 The Inner Detector 26
2.3.5 Calorimeters 27
2.3.6 Muon spectrometer 29
2.3.7 Trigger system 31
2.3.8 The ATLAS Simulation Infrastructure 32

3 Hadron reconstruction in ATLAS **33**
3.1 Tracks reconstruction in ID 33
3.2 Jets reconstruction 34
3.2.1 Jet constituents reconstruction 34

3.2.2 Constituents combination and jets formation 37
3.2.3 Jet calibration 37
3.3 Jet substructure variables 38
3.3.1 N-Subjettiness (τ_N) 39
3.3.2 Energy Correlation Functions (ECFs) and D_2 39

4 Anomaly Detection **41**
4.1 What is Anomaly detection? 41
4.2 Deep Learning 42
4.2.1 Mathematical definition 42
4.2.2 Loss function 43
4.2.3 Learning paradigms 44
4.2.4 The steps for a model to learn 44
4.2.5 Estimating the model's performance 45
4.3 Anomaly Detection in High Energy Physics 45
4.4 Towards the statistical analysis 47

5 ATLAS analysis with AD in hadronic final states **49**
5.1 The dataset 49
5.1.1 Final states 50
5.1.2 Signal processes 50
5.1.3 Background processes 51
5.2 Machine Learning 52
5.3 Graph Neural Networks 52
5.4 Anomaly Detection Implementation with Deep Learning 52
5.5 Overview of Anomaly Detection 53
5.6 Graph Neural Networks 54
5.7 Anomaly Detection Implementation via Deep Architectures 54
5.8 CURCH 55
5.9 How is it implemented? 55

6 Conclusions **56**

Bibliography **57**

Introduction

It was 4 July 2012 when Joseph Incandela, CMS spokesperson, affirmed: "*They combine to give us a combined significance of five standard deviations*". With these exact words, he announced, confirmed by ATLAS spokesperson Fabiola Gianotti a few minutes later, the first observation of the Higgs boson [1, 2], which completed the Standard Model (SM) framework. Fourteen years later, no comparable discovery has followed within the SM paradigm at collider experiments, strengthening the hypothesis that it is the most complete and well-tested description of nature.
In reality, the SM probably is not as complete as it appears. In fact, there are many phenomena not included in the framework but confirmed by astrophysical and cosmological observations: among these, we can find the neutrino mass measurement, Dark Matter [3] and baryon asymmetry. All these processes fall into the category of New Physics (NP) Beyond the Standard Model (BSM). They have been hunted by ATLAS and CMS experiments using traditional data analysis techniques, targeting specific theoretical models but, unfortunately, nothing has been observed yet and only upper limits on production cross-sections were set. There are many reasons for this unexpected outcome: wrong physical model for experiments, wrong triggering while selecting data, or deviations too small to be observed through normal techniques.
A possible solution to some of these issues could be Anomaly Detection (AD), that is the search for data deviating from what is already known. In our field of interest, Particle Physics, we apply this method to look for anomalies (signal data), deviating from well known reacting and decaying particles (background data). The strength of this method lies in its model agnostic nature, meaning that it does not rely on a specific theoretical model to analyse data. This allows us to explore a more vast range of events, increasing the background rejection efficiency, giving us a first hint of the region where to look more in depth for New Physics with precise data analysis techniques. In this process, unsupervised Machine Learning (ML) helps speeding up everything.
This thesis investigates whether such an unsupervised approach can be applied to a concrete search. Using ATLAS official Monte Carlo (MC) samples simulating the ATLAS detector during Run-3, a search for BSM resonant signals decaying into hadronic final states with large-radius jets (used to probe boosted resonance topologies) has been conducted, preparing the necessary tools to finalise the analysis on real ATLAS data. Leveraging the Anomaly Score (AS) distributions computed by a Graph Neural Network (GNN) for each jet, the analysis strategy was defined, starting with the definition of Working Points (WPs), thresholds on the AS distribution corresponding to a given background rejection fraction. A crucial part of

the strategy was determining the correct modelling function for the background and signals di-jet invariant mass (m_{JJ}) distribution. After tuning and validating a Double-Sided Crystal Ball (DSCB) function to model the signals and a 5-parameter function for the background distribution, fits to the MC samples were executed, deriving the expected upper limits at 95% Confidence Level on m_{JJ} using the CLs method, at different AS WPs. The study on MC samples was concluded and fully validated by performing sensitivity studies based on a binned maximum likelihood fit to the di-jet invariant mass distribution.
The thesis is structured as follows. In Chapter 1 the Standard Model is presented, together with theories of Physics beyond it. The LHC collider and the ATLAS experiment are discussed in Chapter 2. Chapter 4 presents the Anomaly Detection approach. The typical workflow of a Particle Physics analysis is described in Chapter **??**. Chapter 5 presents the analysis conducted during this study and its results. Finally, conclusions are drawn in Chapter 6.

First index draft:

- ~~Introduction~~
- ~~SM & BSM~~
  - ~~Intro SM~~
  - ~~Particles~~
  - ~~QED~~
  - ~~QCD~~
  - ~~EW~~
  - ~~Higgs~~
  - ~~BSM~~
- ~~ATLAS~~
  - ~~LHC accelerator system~~
  - ~~pile-up, EM and hadron showers~~
  - ~~ATLAS detector~~
- ~~Hadron reconstruction in ATLAS~~
- Anomaly Detection
  - What is it?
  - How is it implemented using DNNs?
  - Why AD in our case? How are we implementing it?
- How is a Particle Physics Analysis conducted?
  - MC sample
  - distributions

  - ...
  - CLs method

- ADJJ analysis: plots and results

Chapter 1

(Beyond) The Standard Model

The Standard Model (SM) of Particle Physics is a relativistic Quantum Field Theory (QFT) describing three out of four fundamental interactions, strong, weak and electromagnetic (EM), and the behavior of elementary particles under them. For a long time attempts were made to to merge the fourth fundamental interaction, gravity, with the SM framework, but those have not been validated yet by experimental observations. Nevertheless, for our purposes, gravity does not play a significant role since the energy scales involved in the processes at a collider experiment, as the LHC, are much lower than the Planck scale, where gravitational effects become relevant.
What is remarkable about the SM is that it is a very well tested theory, with predictions confirmed by experimental observations with a very high precision. The discovery of the Higgs boson in 2012 at the LHC, which completed the SM framework, is a clear example of this.
Despite the success of the SM, there are many phenomena that are not explained by it, but are confirmed by astrophysical and cosmological observations, opening the road for Physics Beyond the Standard Model.
This chapter presents a brief overview of the SM, highlighting its major features. In addition, hints for Physics Beyond it are discussed.

1.1 An overview of the Standard Model

The fundamental constituents of the Universe are precisely described by the SM. They are called elementary particles, which are mathematically described as quantum fields $\psi(x)$ and their dynamics are modelled by Lagrangian densities $\mathcal{L}(\psi, \partial_\mu \psi)$ [4]. Such particles can be graphically pictured as presented in Fig. 1.1. They are divided into two fundamental classes based on their spin quantum number. The first class is composed of *fermions*, which cover the first three column of Fig. 1.1 and are organised in three *generations* of increasing proper mass; they have spin 1/2 and constitute *matter*. The second class is that of *bosons*, with integer spin, which *mediate* fundamental interactions and generate mass.
Fermions are further classified into two distinct categories depending on their participation in the strong interaction. The first category is relative to *quarks*. They carry *colour charge* (red, green, blue), making them triplets under $SU(3)$ and therefore interact via all fundamental forces, among which the strong one. Quarks

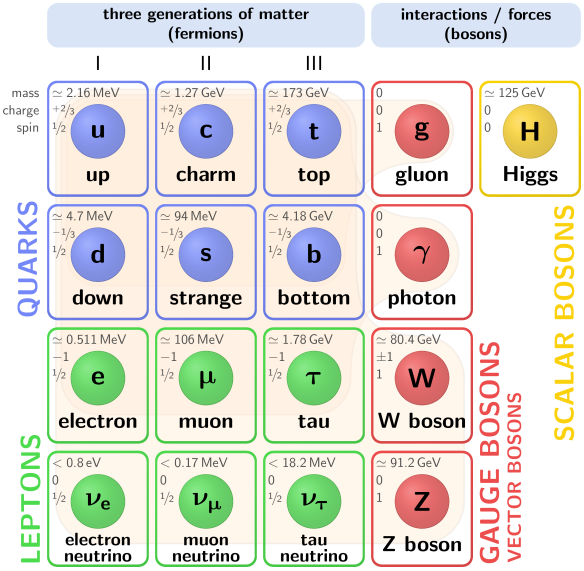


Figure 1.1 The SM framework. Figure adapted from Ref. [5].

exist in six flavors, organized into three generations of weak SU(2) doublets:

$$\begin{pmatrix} u \\ d \end{pmatrix}, \quad \begin{pmatrix} c \\ s \end{pmatrix}, \quad \begin{pmatrix} t \\ b \end{pmatrix} \tag{1.1}$$

Quarks carry a fractional electromagnetic charge: the *up-type* ones (u, c, t) have a charge of $+2/3$, while the *down-type* ones (d, s, b) carry a $-1/3$ charge.
The second category of fermions is associated with *leptons*; there are six of them and similarly grouped into three generations:

$$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix}, \quad \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix}, \quad \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix} \tag{1.2}$$

The charged leptons (e^-, μ^-, τ^-) carry an electric charge of -1 and interact via electromagnetic and weak forces. Neutrinos $(\nu_e, \nu_\mu, \nu_\tau)$ are electrically neutral and interact exclusively via the weak force.
Both quarks and charged leptons, have their own anti-particles, which have the same mass as the matter components but opposite quantum numbers. Regarding neutrinos, it is still debated and to be proved whether they have a distinct anti-particle or they are Majorana fermions, being their own anti-particle.
The second class of particles is associated with mediatiors and mass. Spin 1 vector bosons, known as *gauge bosons*, are the mediators of three of the fundamental interactions, which arise naturally from the requirement of local gauge invariance. Among those fields we find the 8 massless gluons g mediating the strong interaction, the massless photon γ is associated to electromagnetic interaction and 3 massive vector bosons, $W^\pm$ and Z^0, with masses $m_W = 80.36$ GeV, $m_Z = 91.19$ GeV, mediating the weak interaction.
In conclusion, a spin 0 scalar particle, the *Higgs boson* H, with a mass $m_H = 125.13$ GeV, completes the particle spectrum, arising as a physical state from the

spontaneous breaking of electroweak symmetry.

The SM is a gauge theory, meaning that the Lagrangian is invariant under local transformations of the fields. The Lagrangian density describing the SM dynamics can be written as:

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}\gamma^{\mu}D_{\mu}\psi + (D_{\mu}\phi)^{\dagger}(D^{\mu}\phi) - V(\phi) + \bar{\psi}_i y_{ij}\psi_j\phi + h.c., \quad (1.3)$$

The gauge group of the SM, under which such Lagrangian is invariant, is a non-abelian symmetry group defined as:

$$SU(3)_C \otimes SU(2)_L \otimes U(1)_Y \quad (1.4)$$

where $SU(3)_C$ describes the strong interaction via Quantum Chromodynamics (QCD) with subscript C standing for colour, and $SU(2)_L \otimes U(1)_Y$ represents the unified electroweak (EW) interaction, with L associated to the left component of fermions and Y to hypercharge.
The terms of the equation 1.3 above are presented in detail in the following sections.

1.1.1 Quantum Electrodynamics

The first and simplest gauge theory within the SM framework is Quantum Electrodynamics (QED). It describes the electromagnetic interactions between a charged massive fermion, with mass m and represented by a field $\psi(x)$, and the photon, the massless gauge boson represented by the vector field A_μ. Using the Lagrangian formalism [6], QED is constructed starting from the free Dirac Lagrangian for a massive fermion [7]:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi \quad (1.5)$$

In the equation above, $\bar{\psi} = \psi^{\dagger}\gamma^0$ is the Dirac adjoint of the field ψ and γ^μ are the Dirac matrices derived from Pauli matrices σ_i (i =1,2,3 and μ=0,1,2,3):

$$\gamma^0 = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \quad (1.6)$$

The Dirac Lagrangian 1.5 is invariant under a global $U(1)$ phase transformation $\psi(x) \rightarrow e^{ie\alpha}\psi(x)$, with e the electric charge of the fermion and α a constant phase. Such invariance is not preserved under a local $U(1)$ phase transformation $\psi(x) \rightarrow e^{ie\alpha(x)}\psi(x)$, where $\alpha(x)$ is a space-time dependent phase. In order to preserve the invariance, the partial derivative ∂_μ is replaced by the covariant derivative D_μ:

$$D_{\mu} = \partial_{\mu} + ieA_{\mu}(x) \quad (1.7)$$

where $A_\mu(x)$ is a vector gauge field representing the photon field, which transforms as $A_\mu(x) \rightarrow A_\mu(x) - \partial_\mu\alpha(x)$. Including the gauge-invariant kinetic term to take into account for the new field A_μ, we derive the full QED Lagrangian:

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^{\mu}D_{\mu} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad (1.8)$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor. The interaction term $-e\bar{\psi}\gamma^\mu\psi A_\mu$ rules the coupling of charged fermions to photons with a strength proportional to the electric charge e.
The invariance under a local $U(1)$ transformation gurantees, via the Noether's theorem [8], the existence of a conserved current and charge which, in the case of QED, are the electromagnetic current and the electric charge, respectively.

1.1.2 Quantum Chromodynamics

The building of the Lagrangian for QCD is similar to the one for QED: the Dirac Lagrangian is required to be invariant under a local symmetry group, which in this case is the non-Abelian $SU(3)_C$ group, where C represents the triplet color degree of freedom (red, green, blue), the conserved charge of QCD [9]. This theory models the strong interaction between *quarks*, fundamental constituents of hadrons represented as fermionic fields $q(x)$, and *gluons*, the massless gauge bosons represented as vector fields $G^a_{\mu\nu}(x)$ with $a = 1, ..., 8$. Similarly to QED, the local transformation of the quark field is defined as $q(x) \to q'(x) = e^{ig_s T^a \alpha_a(x)} q(x)$, with g_s the strong coupling constant and $\alpha_a(x)$ the local phase. The term T^a appearing in the previous transformation equation is associated to the generators of the $SU(3)_C$ group, which can be expressed via the Gell-Mann matrices as $T^a = \frac{1}{2}\lambda^a$; such generators satisfy $[T^a, T^a] = if^{abc}T^c$ with f^{abc} being the structure constants of $SU(3)$.
Also in this case, a covariant derivative is introduced to preserve the invariance of the Lagrangian under local transformations:

$$D_\mu = \partial_\mu + ig_s T_a G^a_\mu \tag{1.9}$$

The non-Abelian structure of the group, gives rise to the QCD field tensor:

$$F^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu - g_s f^{abc} G^b_\mu G^c_\nu \tag{1.10}$$

The presence of the non-linear structure factor f^{abc} both in the commutators and in the tensor, is associated to a particular feature of QCD: the triple and quadruple gluon self-interaction of gluons, a feature not present in QED.
Finally, the Lagrangian describing the dynamics of QCD is written as:

$$\mathcal{L}_{\text{QCD}} = \bar{q}\left(i\gamma^\mu D_\mu - m\right) q - \frac{1}{4} F^a_{\mu\nu} F^{a,\mu\nu} \tag{1.11}$$

The term g_s, cited before in the local transformation, is usually presented as $\alpha_s = \frac{g_s^2}{4\pi}$, which is the running coupling constant of QCD, whose value depends on the energy scale Q^2 of the process. The non-Abelian nature of QCD gives rise to a peculiar behavior of $\alpha_s(Q^2)$. In fact, at high energy scales (meaning short distances), $\alpha_s(Q^2)$ decreases logarithmically, giving rise to the *asymptotic freedom*, enabling accurate calculations via perturbative QCD (pQCD). On the other hand, at low energy scales (meaning long distances), $\alpha_s(Q^2)$ increases, giving rise to the *color confinement*, which prevents colored particles from existing as isolated free states. In this regime, quarks and gluons undergo fragmentation and hadronization, producing collimated sprays of energetic hadrons, known as *jets*.

1.1.3 Electroweak model

Making use of the Glashow-Weinberg-Salam (GWS) model [7, 10], it is possible to derive a unified description of *electromagnetic* interaction, already presented in Sec. 1.1.1, and the final third interaction, the *weak* one.
To derive the Electroweak (EW) Lagrangian, the steps to follow are similar to those already presented for QED and QCD. In the first place, the symmetry group under which the Lagrangian is invariant is defined, which is $SU(2)_L \otimes U(1)_Y$ in this case, with Y known as hypercharge, the quantum number of EW interactions.
Secondly, the description of fermions and gauge bosons is defined. Concerning the latter, we have four bosons: W_μ^i with $i = 1, 2, 3$ associated to $SU(2)_L$ and B_μ associated to $U(1)_Y$, with coupling constants g and g' respectively. Moving to fermions, they are chirality eigenstates. This means that they can be expressed as left-handed and right-handed components, which are defined as $\psi_{L,R} = \frac{1}{2}(1 \mp \gamma^5)\psi$, with $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. The left-handed components of fermions transform as doublets under $SU(2)_L$ and are described as $L = (\nu, l^-)$ for leptons and $Q = (u, d)$ for quarks. They have quantum numbers $(T = 1/2, T_3 = \pm 1/2)$, with T being the isospin. The right-handed components, instead, transform as singlets under $U(1)_Y$, with quantum numbers $(T = 0, T_3 = 0)$.
The transformation of the fermion components is defined in a compact way as:

$$\begin{aligned} \psi_L(x) &\longrightarrow \psi'_L(x) = e^{i\vec{\alpha}(x)\cdot\vec{\tau} + i\beta(x)Y}\psi_L(x), \\ \psi_R(x) &\longrightarrow \psi'_R(x) = e^{i\beta(x)Y}\psi_R(x) \end{aligned} \tag{1.12}$$

where $\vec{\tau} = \frac{1}{2}\vec{\sigma}$ are the weak isospin, generators of $SU(2)_L$.
Electric charge Q, weak isospin T_3, and weak hypercharge Y are connected by the relation:

$$Q = T_3 + \frac{Y}{2} \tag{1.13}$$

The following step towards the EW Lagrangian, is to introduce the covariant derivative to preserve the invariance of the Lagrangian under local transformations. The derivative is defined as:

$$D_\mu = \partial_\mu + ig\frac{\tau^a}{2}W_\mu^a(x) + ig'\frac{Y}{2}B_\mu(x) \tag{1.14}$$

At this point, the EW Lagrangian can be written as:

$$\mathcal{L}_{\text{EW}} = -\frac{1}{4}W^{a,\mu\nu}W^a_{\mu\nu} - \frac{1}{4}B^{\mu\nu}B_{\mu\nu} + i\bar{\psi}\gamma^\mu D_\mu\psi \tag{1.15}$$

where $W^{a,\mu\nu}$ and $B^{\mu\nu}$ are the field strength tensors associated to the gauge bosons W_μ^a and B_μ, respectively.
In conclusion, if a rotaion is applied to the W_μ^a and B_μ fields, the four physical gauge bosons $\gamma, Z, W^\pm$ are obtained as:

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix}\begin{pmatrix} B_\mu \\ W_\mu^3 \end{pmatrix}, \quad W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2) \tag{1.16}$$

The parameter appearing in the previous equation is the Weinberg angle θ_W, which is defined as $\tan\theta_W = g'/g$.

The EW Lagrangian 1.15 does not provide a proper description of the mass of the bosons involved, which is in contrast with the experimental observations of massive $W^{\pm}$ and Z^0 bosons. The solution to this issue is provided by the Brout-Englert-Higgs mechanism (BEH), which is presented in the following section.

1.1.4 The Brout-Englert-Higgs mechanism

The BEH mechanism provides a theoretical framework to generate masses for the gauge bosons and fermions of the SM, while preserving the local gauge invariance of the Lagrangian. The mechanism is based on the concept of spontaneous symmetry breaking (SSB) of the group $SU(3)_C \otimes SU(2)_L \otimes U(1)_Y$ into $SU(2)_L \otimes U(1)_{EM}$ [11]. The SSB is realized by introducing a complex scalar field ϕ, which is a doublet under $SU(2)_L$ and carries hypercharge $Y = 1/2$. It is defined as:

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi_1 + i\phi_2 \\ \phi_3 + i\phi_4 \end{pmatrix} \tag{1.17}$$

The potential associated to the scalar field is defined as:

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \frac{\lambda}{2} \left(\phi^\dagger \phi\right)^2 \tag{1.18}$$

where μ^2 and λ are constants. Among the two constants, λ must be positive to ensure the stability of the potential, while μ^2 can be either positive or negative. The shape of the potential is determined by the sign of μ^2. If $-\mu^2 > 0$, the potential has a single minimum at $\phi = 0$, preserving the symmetry. If $-\mu^2 < 0$, the potential gets the well-known mexican hat shape presented in Fig 1.2.

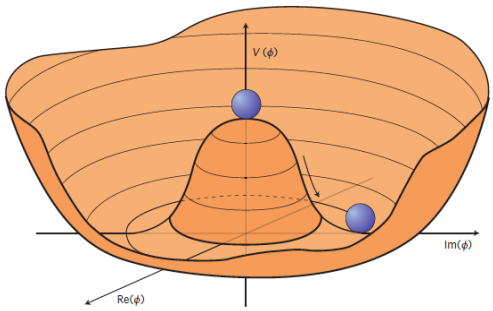


Figure 1.2 Representation of the Higgs potential $V(\phi)$. Figure from Ref. [12].

In the latter case, the vacuum state, as known as the minimum of the potential, is degenerate and associated to a non-zero value of the scalar field ϕ. In particular, the vacuum expectation value (VEV) v of the Higgs field, defined as $v = \sqrt{\mu^2/\lambda}$, is not null. As a direct consqeuence, the global $U(1)_Y$ symmetry is spontaneously broken. The VEV is given by:

$$\phi^\dagger \phi = \frac{\mu^2}{2\lambda} = \frac{v^2}{2}, \quad \phi_0 = \frac{v}{\sqrt{2}} \tag{1.19}$$

Since the theory is built to be gauge invariant, in order to remove the unphysical degrees of freedom, the unitary gauge transformation is applied to the scalar field ϕ.

In this gauge, the Higgs field can be expressed as:

$$\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix} \tag{1.20}$$

where $h(x)$ is the Higgs boson scalar field.
Using the definition of ϕ presented in Eq. 1.20, the EW Lagrangian density can be written as:

$$\begin{aligned} &\mathcal{L}_{\text{SM}} = -\tfrac{1}{4} W^{a,\mu\nu} W^{a}_{\mu\nu} - \tfrac{1}{4} B^{\mu\nu} B_{\mu\nu} + i\overline{\psi}\gamma^{\mu} D_{\mu}\psi + \\ &-Y^{l}_{ij}\overline{L_i}\phi\, e_j - Y^{\nu}_{ij}\overline{e_i}\phi\, L_j - Y^{d}_{ij}\overline{Q_i}\phi\, d'_j - Y^{u}_{ij}\overline{Q_i}\phi\, u_j + \text{h.c.} \\ &+(D^{\mu}\phi)^{\dagger}(D_{\mu}\phi) - V(\phi) \end{aligned} \tag{1.21}$$

where the term D_μ is the covariant derivative defined in Eq. 1.14 and Y_{ij} are the Yukawa couplings. Each line of the Lagrangian density 1.21 has as specific role. The first line represents the gauge bosons and how they interact with fermions, the second one contains the Yukawa couplings which is associated to Higgs coupling with fermions, and the third line contains the kinetic term for the Higgs field and its potential.
The Higgs potential $V(\phi)$, defined in Eq. 1.18, with che chosen gauge, can be expressed as:

$$V(h) = \frac{1}{2} m_h^2 h^2 + \frac{m_h^2}{2v} h^3 + \frac{m_h^2}{8v^2} h^4 \tag{1.22}$$

The m_h term appearing in the previous equation is the mass of the Higgs boson, which is a free parameter of the SM, depending on the parameters of the potential, as $m_h = \sqrt{2\lambda v^2} = \sqrt{2\mu^2}$. The cubic and quartic terms in h represent self-interactions of the Higgs boson.
The Higgs boson mass has been measured by the ATLAS and CMS collaborations [1, 2] to be $m_h = (125.13 \pm 0.11)$ GeV, which is consistent with the SM predictions. From this measurement, the value of VEV v can be derived as $v \approx 246$ GeV.

1.2 Hints of Physics Beyond the Standard Model

The SM presented in the previous section 1.1 is a very successful theory, which has been confirmed by experimental observations with a very high precision. Despite its success, the SM probably is not a complete theory, given the many open questions that are not answered by the SM and motivate the search for Physics Beyond the Standard Model (BSM).
Among these questions, we find the nature of the Higgs boson, whether it is the only scalar particle of this type or it is part of an extended Higgs sector. In fact, many extensions of the SM predict the existence of additional scalar or pseudo-scalar states, such as two-Higgs-doublet models (2HDMs), which are currently studied at the LHC by the ATLAS collaboration [13].
Together with theories to be proven, as the one just cited, there are many other already proven by astrophysical and cosmological observations. The first one is the existence of Dark Matter (DM), which is known to compose 26.4% of the Universe,

together with a 68.7% of Dark Energy and a 4.9% of ordinary matter [14]. The presence of DM is inferred from its gravitational effects on visible matter, such as galaxies rotation curves [15], galaxy-galaxy gravitational lensing [16] and the Bullet Cluster [17], the result of the collision of two galaxy clusters. In order to observe DM at collider experiments, the approach, known as indirect search, is to detect decay products of DM particles. This type of search was not able to provide any evidence of DM particles yet, leaving it an ongoing study at the LHC.
Another open question is the matter-antimatter asymmetry in the Universe, also known as baryon asymmetry. In fact, it is agreed that in the early Universe there was balance between matter and antimatter, but today in the Universe, after its cooling down, the balance is slightly broken in favour of matter. Such unbalance could be associated with a violation of the CP symmetry, which is a combination of charge conjugation C and parity P. This motivates the search for new sources of CP violation, which could be provided by BSM theories.
One final question, among many, is the hierarchy problem, which is associated to the Higgs boson mass. In fact, the Higgs boson mass value is subject to quantum loop corrections, which are quadratically divergent. This means that the Higgs boson mass receives contributions from all energies up to the Planck scale (10^{19} GeV). Since the Higgs boson mass has been measured to be 125 GeV, an unnatural fine-tuning of the parameters of the SM would be required with large cancellations between different terms in radiative correction. This motivates the search for New Physics (NP) at the TeV scale, which could provide a natural explanation for the Higgs boson mass.

1.2.1 New Physics at colliders

After reading the BSM cases presented above, and many others not cited, it appears to be quite clear that New Physics Beyond the Standard Model should exist and, in principle, it should be accessible at the energy scales probed by the LHC. Given the lack of observations at CERN experiments, it is worth analysing the approach used to probe NP. This kind of experiments are conducted with two main different approaches: *precision measurements*, which are sensitive to small deviations from the SM predictions, and *direct searches*, which aim to produce new particles in high-energy collisions and probe them. The latter direction has been traditionally pursued by targeting a specific BSM model, which is known as *model-dependent* search. In this case, the search is optimized for a specific model and its parameters. The results of such searches are usually presented as exclusion limits on the space of parameters of the model.
A suitable alternative to model-dependent searches is the *model-independent* or *model-agnostic* approach, which searches for deviations from the SM predictions in a general way, without assuming any specific BSM model. This approach is more flexible and can be sensitive to a wider range of BSM scenarios. This type of search is gaining more and more interest in recent years in the LHC program for Run 3 and following.
A model-agnostic search for NP is the topic of this thesis, with the goal to prove its flexibility and sensitivity. Two possible BSM theories have been chosen as benchmark models to test the strategy, Heavy Vector Triplets (HVT) which could explain new

narrow resonances, and Dark Jets (DJ) which are QCD-like jets produced by dark sectors.

1.2.2 Heavy Vector Triplets

Popular studied BSM scenarios are those presented in Simplified Models [18], which assume that NP are associated with new narrow resonances. Such assumption has the direct consequence of reducing the number of free parameters of the model, making it more predictive and easier to test.
The simplest and most studied Simplified Model is the Heavy Vector Triplet (HVT) model [19], which predicts the existence of electroweakly charged spin 1 bosons with a behavior similar to the SM W and Z bosons. This model introduces in the SM Lagrangian the presence of the real vector field V_μ^a $(a = 1, 2, 3)$, which is a triplet under $SU(2)_W$ and has zero hypercharge. Making a combination of these components, the physical states of the HVT model are obtained as:

$$V_\mu^\pm = \frac{V_\mu^1 \mp i V_\mu^2}{\sqrt{2}}, \quad V_\mu^0 = V_\mu^3 \tag{1.23}$$

The HVT undergo Drell-Yan decays with highest branching ratios with two leptons, two quarks or two bosons in the final state.
ATLAS and CMS collaborations have conducted many analyses to search for HVT decaying mainly in fully hadronic final states, which are the most abundant ones. The results of these searches are usually presented as exclusion limits at 95% confidence level on the mass of the HVT bosons, which are currently excluded in the range $[1.3, 3.1]$ TeV by ATLAS [20] and $[3.3, 3.5]$ TeV by CMS [21]. A typical plot of these kind of searches can be found in Fig. 1.3, showing the result of the ATLAS analysis.

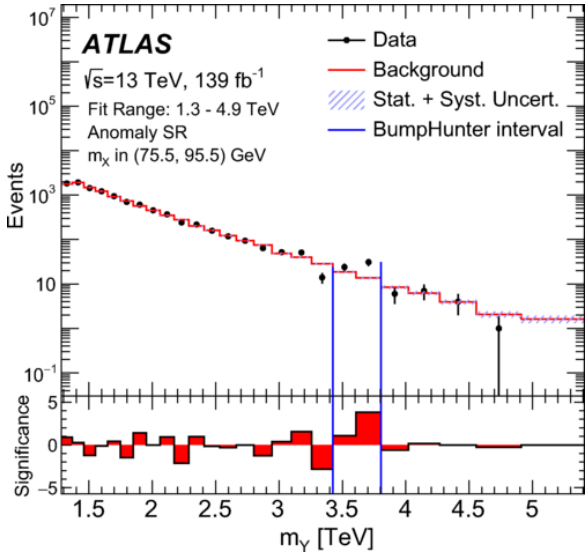


Figure 1.3 BumpHunter scan in the Y boson invariant mass distribution within the HVT framework. Figure from Ref. [20].

1.2.3 Dark Jets

The dark QCD model, a strongly coupled dark sector, has gained interest in the last years as a possible explanation for DM. [22] In this model, the dark sector is composed of dark quarks and gluons, which are charged under a new gauge group $SU(N)_D$ and interact via a new strong force, similar to QCD.
The dark quarks can be pair-produced at colliders via a mediator particle. Once produced, they undergo a shower and hadronization process, similar to QCD, producing collimated sprays of dark hadrons, known as dark jets. This process is predicted to happen at a scale Λ_D, which is the confinement scale of the dark sector, with the assumption that $\Lambda_D << \sqrt{s}$, where $\sqrt{s}$ is the center-of-mass energy of the collision.
The commonly known portal is the mediator particle involved in the interaction between the SM and the dark sector. The experimental signature of those processes at colliders will be jets with missing energy or general features not compatible with QCD.
ATLAS and CMS collaborations have conducted many analyses to search for dark jets. In particular, ATLAS has conducted a search for a $Z' \rightarrow q_D\overline{q}_D$ resonance [23], where q_D is a dark quark and Z' is probed in the range of $[1.5, 3.5]$ TeV. The observed limit on the mass for such new resonance has been set up to 3.5 TeV, as presented in Fig. 1.4.

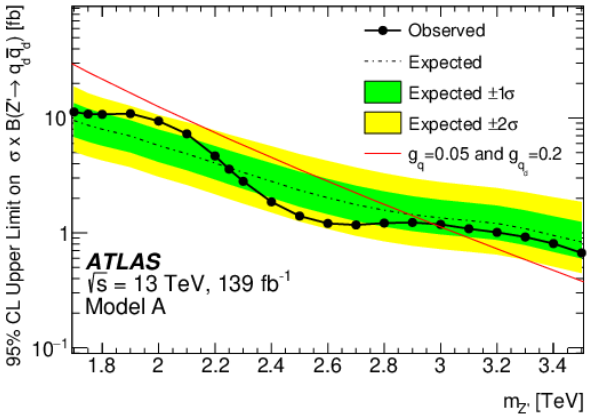


Figure 1.4 Upper limits at 95% CL on the cross section times the branching ratio for a dark QCD process. Figure from Ref. [20].

Chapter 2

The LHC and the ATLAS experiments

The history of the European Organization for Nuclear Research (CERN) traces back to the post-WWII era, when European physicists, including Edoardo Amaldi, Niels Bohr and Pierre Auger, promoted a collaborative international laboratory dedicated to fundamental science. CERN was established in September 1954 and today it hosts the world's largest and most energetic accelerator complex, centered around the Large Hadron Collider (LHC).
In the following, after a description of the LHC machine, the discussion covers the architecture of the ATLAS (A Toroidal LHC ApparatuS) detector, one of the four major experiments at CERN, including its main sub-systems and simulation procedures. A special focus is given to the proton-proton collision dataset collected by ATLAS at a center-of-mass energy of 13.6 TeV during LHC Run 3, which forms the basis for this study.

2.1 The LHC

The Large Hadron Collider (LHC) at CERN, with its 27 kilometres ring [25], is the world's largest particle accelerator. It has four major interaction points which host the major CERN experiments: ATLAS, CMS, LHCb and ALICE. LHC is situated about 100 metres underground for economic[1]and geological reasons, as well as to provide radiation shielding. With its first run on 10 September 2008, it currently accelerates hadrons. Depending on the run, protons or lead nuclei are collided, with a center-of-mass energy of 13.6 TeV and 5.36 TeV respectively. In order to reach such high energies, a complex system of pre-acceleration is adopted, as presented in Fig. 2.1.

[1]Building LHC on the surface would have required the purchase of properties to do so and an under the open sky structure with a 27 km circumference would have been more expensive than agreeing with landowner and excavate a tunnel underground.

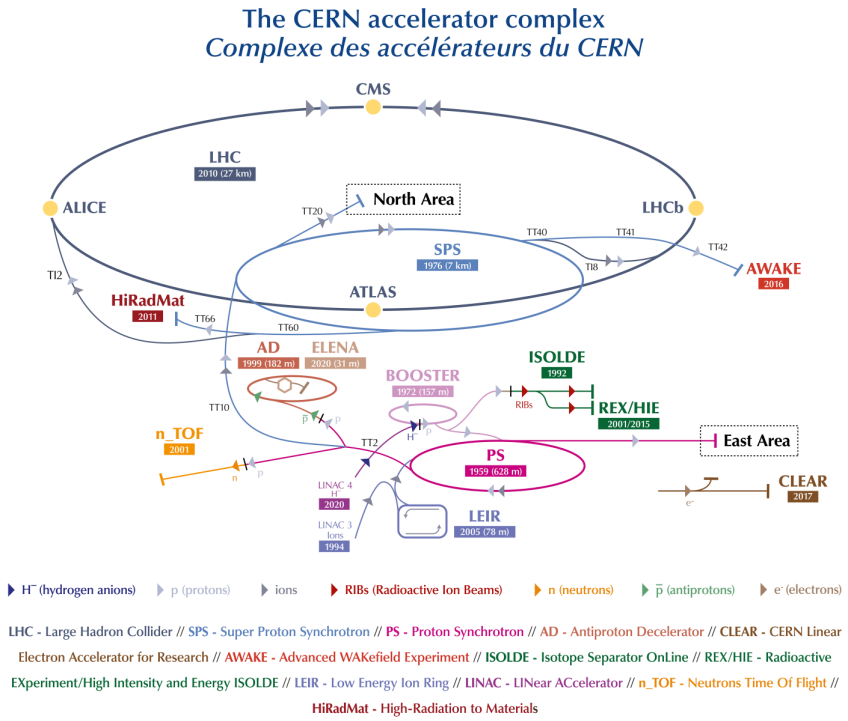


Figure 2.1 CERN accelerating system. Figure from Ref. [24].

2.1.1 CERN's accelerator system

For proton-proton (pp) collisions, the process begins with the extraction of negative hydrogen ions (H^-), which are initially accelerated to an energy of 160 MeV by the Linear Accelerator 4 (LINAC4). Upon injection into the Proton Synchrotron Booster (PSB), the ions are stripped of their two electrons leaving only the bare protons, which are grouped into bunches and boosted to an energy of 2 GeV. The protons then enter the Proton Synchrotron (PS), reaching an energy of 25 GeV. The final step takes place in the Super Proton Synchrotron (SPS) that accelerates the proton bunches up to the LHC injection energy of 450 GeV. Finally, the proton bunches are injected into the two counter-rotating beam pipes of the LHC, where they are accelerated to their maximum operational energy of 6.8 TeV per beam.
In heavy-ion operations (primarily based on lead nuclei, ^{208}Pb), ions are initially accelerated by the Linear Accelerator 3 (LINAC3) to an energy of 4.2 MeV per nucleon. The nucleons are then accumulated and boosted to 72 MeV per nucleon inside the Low Energy Ion Ring (LEIR). In the next stage, the PS accelerates ions to 5.9 GeV per nucleon, before transferring them to the SPS where they reach an energy of 177 GeV per nucleon. When injected into the main LHC ring, the heavy-ion beams are accelerated to a final energy of 2.68 TeV per nucleon.
Particles are accelerated up to such high energies by means of radio-frequency (RF) cavities, resonant structures generating an oscillating electric field, with a fixed frequency of 400MHz, which accelerates charged particles each time they pass through. Particles with the target energy passing trough the cavity while the field is zero,

will not be accelerated, while more (less) energetic particles will be decelerated (accelerated). Using this technique, it is ensured that particles stays in the cited bunches.

2.1.2 The LHC parameters

The LHC is composed of two circular beam pipes, denominated Beam 1 and Beam 2. Inside them, two particle beams, composed of a maximum of 2808 bunches with approximately 1.15×10^{11} protons per bunch and spaced by 25 ns, travel separately close to the speed of light in opposite directions. There are eight Interaction Points (IPs) along the LHC ring, but only four of them are used for collisions, hosting the main LHC experiments: ATLAS at IP1, ALICE (A Large Ion Collider Experiment) at IP2, CMS (Compact Muon Solenoid) at IP5 and LHCb (LHC beauty) at IP8. The beam pipes and the interaction points are presented in Fig. 2.2.

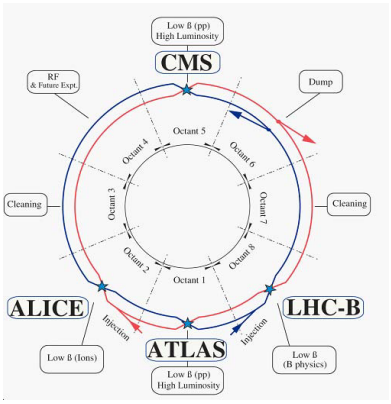


Figure 2.2 Schematic layout of the LHC (Beam 1 clockwise, Beam 2 anticlockwise). Figure from Ref. [26].

Due to the very high speed, when the two beams collide, the energy of the center of mass, known as $\sqrt{s}$, is twice the energy of each beam (which is the same). In fact, knowing that $E^2 = m^2c^4 + p^2c^2$ and using the natural unit system ([m]=eV/c^2, [p]=eV/c), we get: $E^2 = m^2 + p^2 \simeq p^2$, where at the end an ultra-relativistic particle (p»m) was assumed. This information is used in the norm of the four-momentum of two colliding particles:

$$\sqrt{s} = \sqrt{P_\mu P^\mu} = \sqrt{E_{tot}^2 - |\vec{p_{tot}}|^2} = \sqrt{(E_1 + E_2)^2 - |\vec{p_1} - \vec{p_2}|^2} \simeq 2E \qquad (2.1)$$

where we assumed $E_1 = E_2$. The 6.8 TeV of energy reached by LHC protons confirm the assumption p»m and it leads to the known value of $\sqrt{s} = 13.6$ TeV.
Accelerated beams are directed and curved by superconducting electromagnets with a magnetic field of up to 8.3 T [27]: 1232 dipole magnets and 392 quadrupole magnets are used to maintain the beams focused. Magnets are kept at a temperature of 1.9 K to ensure a 11600 A electric current which is needed for the intense magnetic field. Despite their essential contribution, magnets also represent one of the main limitations of this apparatus. In fact, from the Lorentz force equation $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$, it is possible to get a connection between the external magnetic

field $\vec{B}$ (the one generated by superconducting magnets), the momentum p of the particle circulating inside the beams and its radius of curvature R, which is:

$$p[GeV] = 0.3B[T]R[m] \tag{2.2}$$

It is evident from Eq. 2.2 that, to keep a constant momentum p of the particles circulating inside the beams, a higher magnetic field B is required in case of reduction of the radius of curvature R. Unfortunately, current technologies set a limit to the magnetic field generated by a magnet, limiting also the minum dimension of an accelerator ring.
Performance of the LHC is evaluated based on an essential parameter: the instantaneous luminosity $\mathcal{L}$. It defines the number of collisions per cm^2 and per second produced in the collider and thus it is measured in $cm^{-2}s^{-1}$. It is computed as follows [28]:

$$\mathcal{L} = \frac{N_b^2 n_b f_{rev} \gamma}{4\pi\epsilon_n \beta^*} F \tag{2.3}$$

where: N_b is the number of particles in a bunch, n_b is the number of bunches in a beam, ϵ_n is the normalized transverse beam emittance, β^* is the beta function at the collision point, γ is the Lorentz relativistic factor computed as $\gamma = 1/\sqrt{1 - v^2/c^2}$, f_{rev} is the revolution frequency. There is a connection between the beta function and the transverse beam size σ, which is $\sigma = \sqrt{\epsilon_n \beta^*/\gamma}$. F represents the geometric luminosity reduction factor:

$$F = \frac{1}{\sqrt{1 + \left(\frac{\theta_c \sigma_z}{2\sigma^*}\right)^2}} \tag{2.4}$$

where θ_c is the full crossing angle at the interaction point, σ_z is the RMS bunch length and σ^* is the transverse RMS beam size at the interaction point. When the instantaneous luminosity goes below a fixed threshold during operations, beams are dumped and a new fill of the LHC is performed.
During LHC Run 3, $\mathcal{L}$ gets the value of 10^{34} cm$^{-2}s^{-1}$ for both ATLAS and CMS experiments, while for LHCb and ALICE it is 10^{32} cm$^{-2}s^{-1}$ and 10^{27} cm$^{-2}s^{-1}$ respectively.
The instantaneous luminosity is linked with the number of collisions per second :

$$\frac{dN}{dt} = \mathcal{L}\sigma_{tot} \tag{2.5}$$

where σ_{tot} is the total cross section, which is the measure of the probability of a specific interaction process to happen. If the instantaneous luminosity is integrated over time, the number of events collected is obtained:

$$N_{events} = \sigma_{tot} \int \mathcal{L} dt \tag{2.6}$$

2.1.3 LHC Run Schedule

The LHC operations are structured into extended physics data-taking campaigns named *Runs*, which are separated by planned technical maintenance and machine

upgrade periods, which are known as Long Shutdowns (LS). It was Run 1, from 2010 to 2013, that started the LHC physics program, operating at center-of-mass energies of $\sqrt{s} = 7$ TeV and 8 TeV, with a delivered integrated luminosity of 28 fb^{-1}. Run 1 culminated in the discovery of the Higgs boson by the ATLAS and CMS collaborations. Following upgrades during LS1, meant to enable higher-energy operations, Run 2 took place between 2015 and 2018 at $\sqrt{s} = 13$ TeV, delivering an integrated luminosity of approximately 156 fb^{-1}. After another round of upgrades during LS2 the collider entered the Run 3 era in 2022, which lasted until June 2026, operating at an increased center-of-mass energy of $\sqrt{s} = 13.6$ TeV. After having delivered an integrated luminosity of approximately 351 fb^{-1}, the LHC was turned off for the beginning of LS3. This new stop will be used to prepare the LHC for the High-Luminosity LHC (HL-LHC) era, which is expected to start in 2029 and will last until 2042. The HL-LHC will operate at the same center-of-mass energy of Run 3 and is expected to deliver an integrated luminosity of 3000 fb^{-1} during the future Runs.
A graph showing the LHC delivered integrated luminosity during the different Runs, together with the recorded luminosity by ATLAS, is shown in Fig. 2.3.

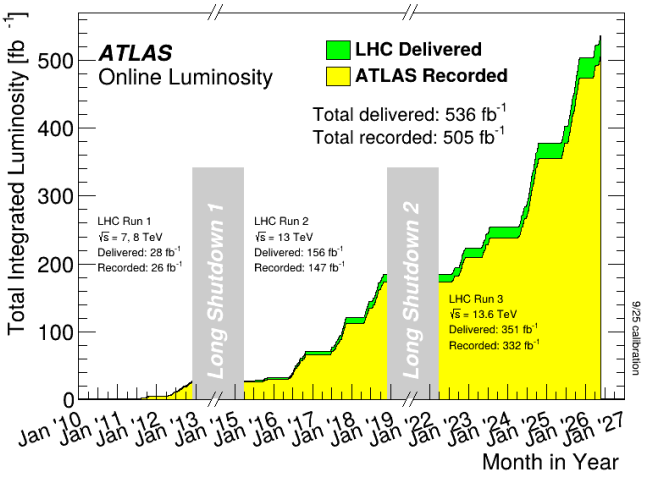


Figure 2.3 Cumulative integrated luminosity delivered to and recorded by the ATLAS Experiment during LHC Run 1, 2 and 3. Figure from Ref. [29].

2.2 *pp* collisions at the LHC

As presented in Sec. 2.1.2, at the LHC protons are collided every 25 ns at a center-of-mass energy of $\sqrt{s} = 13.6$ TeV. Theoretical calculations to model such events using QCD are not trivial, since protons are not elementary particles but composite states of quarks and gluons, collectively referred as *partons* [30]. The canonical approach used with other theories, the *perturbation theory*, can still be used in the case of high-energy interactions, known as *hard scattering*, where the strong coupling constant α_s is small enough to allow a perturbative expansion. It fails at low energy scales, in the so called *soft scattering*, where α_s becomes large and the perturbative expansion diverges.

In these cases, the *factorization theorem* allows to compute the total cross-section of *pp* collisions as a convolution of short-distance partonic matrix elements with long-distance Parton Distribution Functions (PDFs), which describe the probability of finding a parton with a given momentum fraction x inside the proton.
The situation described above gets a degree of complexity when considering that, in a single bunch crossing, multiple *pp* collisions can occur. This phenomenon, known as *pile-up*, is discussed in Sec. 2.2.1.
As said, the calculation of the total cross-section of *pp* collisions is not trivial. For this reason, event generators are used to simulate the physics of *pp* collisions, as described in Sec. 2.3.8 for ATLAS.
In Fig. 2.4 a Feynman diagram of a *pp* collision is shown, where the two protons are represented as two bunches of partons, among which only two of them collide and produce the final state.

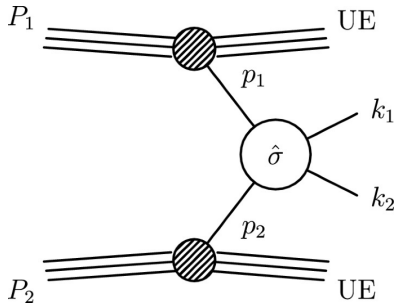


Figure 2.4 Feynman diagram of a *pp* collision. Figure from Ref. [31].

2.2.1 Pile-Up and hadronisation

The partons coming out of the hard scattering process emit gluon radiation and produce quark pairs in a cascading parton shower (Sec. 2.2.3).
When approaching energies compatible with the QCD scale $\Lambda_{\text{QCD}} \approx 200/300$ MeV, the soft scattering regime is reached, with the divergence of the perturbative expansion and the araising of colour confinment. For this reason, colour charged partons must bind into colour-neutral hadrons. This non-perturbative transition is known as *hadronisation* [31]. The processes and collisions not related to the primary hard scatter are defined as *underlying events*, which can rise from multiple sources, such as initial of final state radiation. This type of events, together with soft scattering processes from other partons, are responsible for the Pile-Up (PU) phenomenon. PU degrades event reconstruction and it has to be carefully controlled to ensure a correct physics analysis. When referring to PU, the average number of *pp* collisions per bunch crossing is presented, using the symbol $\langle\mu\rangle$.
In Fig. 2.5, a plot showing the PU distribution in ATLAS from Run 1 to Run 3 is presented. It is evident that the average number of *pp* collisions per bunch crossing has increased over the years, reaching a value of $\langle\mu\rangle \approx 62$ during Run 3.

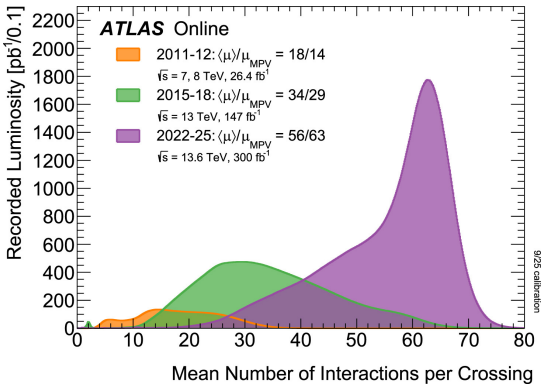


Figure 2.5 Pile-Up distribution in ATLAS from Run 1 to Run 3. Figure from Ref. [32].

2.2.2 Electromagnetic Showers

Before discussing the hadronic showers, which are essential in the formation of jets, the main focus of this thesis, it is important to introduce the electromagnetic version of them. Electromagnetic showers develop when high-energy electrons or photons interact with heavy materials. They are the basis of the electromagnetic calorimeter design, which is used to measure the energy of electrons and photons (Sec. 2.3.5). The passive material, or absorber, is typically a high-Z material with a high probability of interacting with the incoming particles. These interactions reduce particles' energy and slowly stop them.
Electromagnetic showers develop via two primary mechanisms: bremsstrahlung and pair production. Electrons (and positrons) lose energy via bremsstrahlung, which can be defined as the emission of photons (radiation). Those photons undergo pair production, converting them into electron-positron pairs which then emit more photons via bremsstrahlung. This cycle repeats until electrons and photons loose their energies.
The count of particles involved in the cascade grows exponentially, forming a literal *shower* of particles, as shown in Fig 2.6.

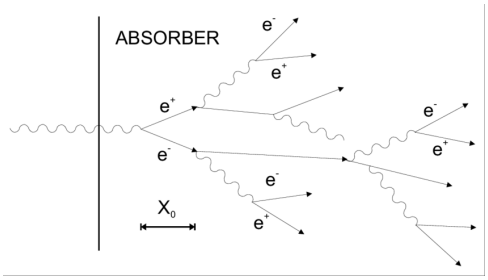


Figure 2.6 Schematic of an electromagnetic shower starting from a photon.

The fundamental parameter of the electromagnetic shower is the radiation length X_0, the mean distance over which an electron loses a factor $1/e$ of its energy. A typical EM shower has a longitudinal profile of $\sim 20\ X_0$. The calorimeter must be deep enough to contain the entire shower and measure the energy of the incoming particle.

2.2.3 Hadronic Showers

Hadronic showers develop when particles subject to the strong interaction collide with nuclei in the calorimeter passive absorber. These inelastic collisions initiate a nuclear cascade with complex modelling. The shower, in fact, contains both an electromagnetic and a hadronic component. Neutral pions (π^0), for instance, decay immediately into photon pairs, initiating the electromagnetic sub-showers presented before. Charged pions ($\pi^{\pm}$), instead, continue to undergo strong interactions, producing secondary hadrons.
The characteristic scale of the hadronic shower development is the interaction length λ, a scale much larger than X_0. Again, the hadronic calorimeter needs to be deep enough to manage the containment of the entire shower.

2.3 The ATLAS detector

A Toroidal LHC ApparatuS (ATLAS) [33] is the world's largest particle detector, situated at Interaction Point 1 at CERN. 44m in length and 25m in height are required to investigate collisions between particles accelerated by the LHC, to study phenomena such dark matter, the Higgs boson and top quark. The ATLAS detector is composed of multiple sub-detectors organized in layers with a cylindrical symmetry around the beam pipe. Among the main sub-detectors we can find: the Inner Detector (ID), the Electromagnetic Calorimeter (ECal), the Hadronic Calorimeter (HCal) and the Muon Spectrometer (MS), as shown in Fig. 2.7.

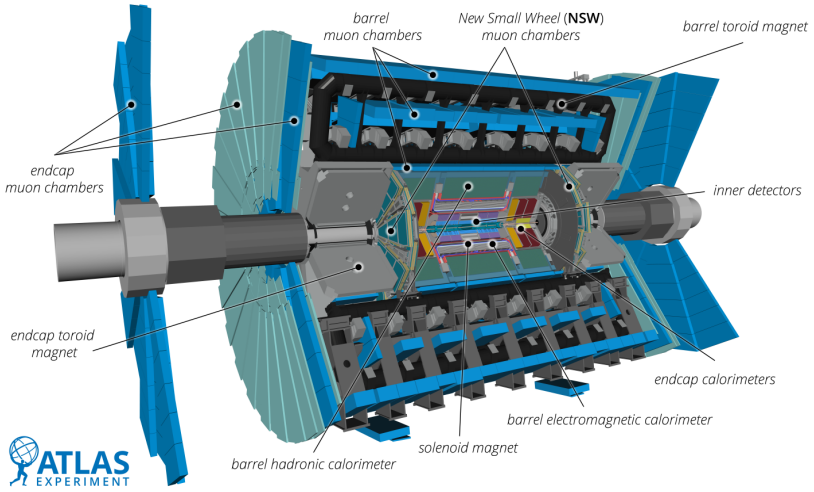


Figure 2.7 ATLAS schematic. Figure form Ref. [34].

2.3.1 Coordinate system

Before analysing the ATLAS detector, it is important to introduce the coordinate system used to describe the position of particles and their momenta inside the sub-detectors. In Fig. 2.8 an overview of the system is presented.

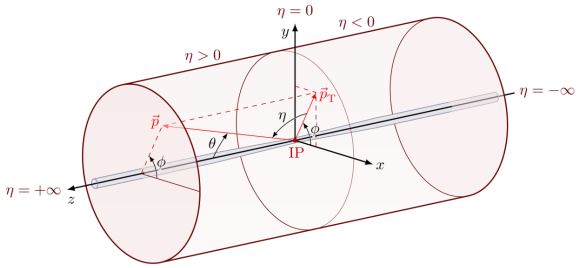


Figure 2.8 Schematic of ATLAS coordinate system

Cartesian axes (*x,y,z*) are constructed as follows. The origin is set in the interaction point (IP). From it, the z-axis departs parallel to the beam tunnel with positive z-values in anticlockwise direction. As a direct consequence, the x-y plane is perpendicular to it, with positive x-values towards the center of the LHC and positive y-values upward. On this Cartesian system, a more natural cylindrical coordinate system (r, θ, ϕ) is defined to describe geometric position and components of particle momenta. r is the absolute distance from the interaction point. ϕ is the azimuth angle, measured around the z-axis, with values $[-\pi, \pi]$. θ is the polar angle, measured starting from the z-axis, with values $[0, \pi]$.
Two more coordinates are defined to describe the position of particles in the detector: the rapidity y and the pseudo-rapidity η. They are defined as:

$$y = \frac{1}{2} ln\left(\frac{E + p_z}{E - p_z}\right) = \text{arctanh}\left(\frac{p_z}{E}\right) \tag{2.7}$$

$$\eta = -ln\left[tan\left(\frac{\theta}{2}\right)\right] = \text{arctanh}\left(\frac{p_z}{|\,\vec{p}\,|}\right) \tag{2.8}$$

where $\vec{p}$ is the three-momentum of the particle and p_z is its component along the z-axis. E represents the energy of the particle. y and η get values in the range $[-\infty, \infty]$, being zero for $\theta = \pi/2$ and $\pm\infty$ for $\theta = 0$ or $\theta = \pi$.
A schematic of the connection between the pseudo-rapidity and the polar angle θ is shown in Fig. 2.9.

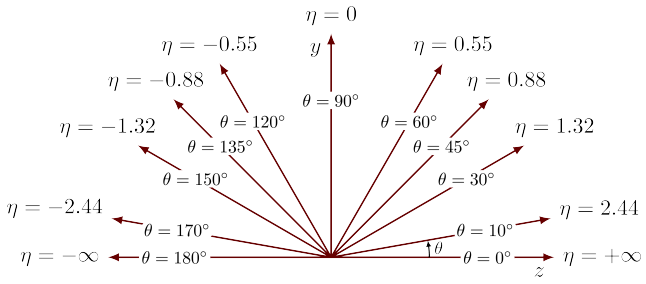


Figure 2.9 Schematic of pseudo-rapidity as a function of the polar angle θ.

These latter two variables are introduced because the difference of rapidities between two particles is a Lorentz invariant quantity under boosts along the z-axis. Since

Lorentz boosts are applied to move from the laboratory frame to the center-of-mass frame of the colliding particles, η and y are widely used instead of the polar angle θ. In case of an ultra-relativistic particle, the rapidity and the pseudo-rapidity are approximately equal, $\eta \approx y$, as shown in Eq. 2.7 and Eq. 2.8. The main difference between the two is that the rapidity depends both on the energy of the particle and its momentum, a combined measurement not trivial at high energies, while the pseudo-rapidity depends only on the particle's momentum. For these reasons, the pseudo-rapidity is preferred to the rapidity, making (r, η, ϕ) the used coordinate system. With such choice of coordinates, the detector components placed at high values of pseudo-rapidity are addressed as *forward region*.
The pseudo-rapidity and the azimuth angle, are used to define a geometrical distance, applied when computing the separation between two physics objects:

$$\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2} \tag{2.9}$$

Prior to an analysis of materials, some more useful quantities are defined. The first one is the transverse momentum p_T, which is the projection of the momentum of the particle in the transverse plane (x, y) of the interaction, defined as:

$$p_T = \sqrt{p_x^2 + p_y^2} \tag{2.10}$$

The importance of this variables relies in the fact that at the LHC collisions happen mostly along the z-axis, generating Lorentz boosts mainly along this axis. This means that the transverse plane is not affected by these boosts, making p_T a Lorentz invariant quantity.
Similarly, the transverse energy E_T is defined as the projection of the energy of the particle in the transverse plane (x, y) of the interaction, defined as:

$$E_T = E \cdot \frac{p_T}{|\vec{p}|} \tag{2.11}$$

A fundamental variable for the purposes of this thesis is the invariant mass m_{inv}, which is associated to the energy of a system of particles. In the case of two particles, using their four-momenta P_1 and P_2, m_{inv} is defined as:

$$m_{inv} = \sqrt{(P_1 + P_2)^2} = \sqrt{(E_1 + E_2)^2 - |\vec{p_1} + \vec{p_2}|^2} \tag{2.12}$$

The invariant mass is a Lorentz invariant quantity, which is used to measure the mass of a resonant state, composed of a particle decaying into two others. In fact, the invariant mass of the system will be equal to the mass of the mother particle.
Based on what has been discussed on Lorentz boosts at the LHC, the sum of the the transverse momenta of the events is expected to be zero, since the initial state of the collision has no transverse momentum. For this reason, if a particle is not detected, as it happens with neutrinos, its transverse momentum, known as missing transverse momentum $\vec{p_T^{miss}}$, can be computed as the opposite of the sum of the transverse momenta of all the detected particles in the event:

$$\vec{p_T^{miss}} = -\sum_i \vec{p}_{T,i} \tag{2.13}$$

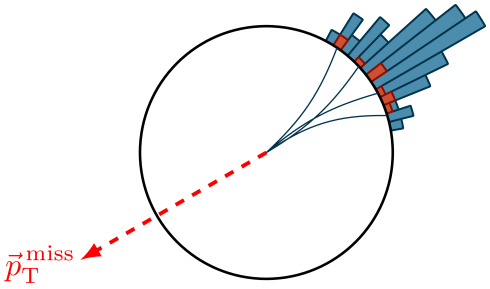


Figure 2.10 Schematic of missing transverse momentum computation.

The computed variable is shown graphically in Fig. 2.10, where the missing transverse momentum is represented as the opposite of the vectorial sum of the transverse momenta of all the detected particles in the event.
In these cases, the transverse mass is used instead of the invariant mass, which is defined as:

$$m_T = \sqrt{2p_T E_T^{miss}[1 - \cos(\Delta\phi)]} \tag{2.14}$$

where E_T^{miss} is the magnitude of the missing transverse momentum and $\Delta\phi$ is the azimuthal angle between the transverse momentum of the detected particle and the missing transverse momentum.

2.3.2 Magnetic system

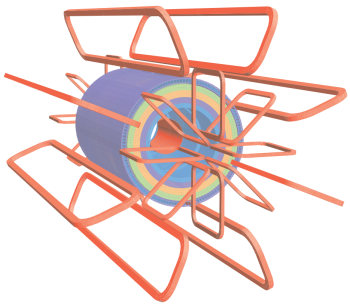


Figure 2.11 ATLAS magnetic system. Figure form Ref. [35].

Magnetic fields play a crucial role in the ALTAS detector, allowing the tracking of charged particles and the measurement of their momenta via the sagitta method [36]. The ATLAS detector relies on a hybrid superconducting magnet system composed of four sub-components: a central solenoid, a barrel toroid and two end-cap toroids [35]. This complex system was adopted to reduce particle interactions with inactive material and, more importantly, to obtain a higher momentum resolution. Cooled to 4.5 K using liquid helium, the system covers a total volume of 26 m in length and 21 m in outer diameter. A schematic of the ATLAS magnetic system is shown in Fig. 2.11.
The central solenoid is positioned inside the calorimetry system directly around the Inner Detector cavity, where it generates a highly uniform 2 T axial magnetic field

parallel to the beam pipe deflecting charged particles in the transverse plane. It measures 5.3 m in length, 19 cm in thickness and an inner diameter of 2.44 m. To minimize dead material in front of the Electromagnetic Calorimeter, the solenoid shares a single cryostat vessel and supportive cylinder with the barrel ECal.
Providing the deflection field for the outer Muon Spectrometer, the barrel toroid is composed of eight large rectangular coils mounted radially with eight-fold azimuthal symmetry around the calorimeters. Extending 25.3 m in length with inner and outer diameters of 9.4 m and 20.1 m, respectively, it generates a peak magnetic field 3.9 T. Two smaller end-cap toroid structures, 5 m long with an inner and an outer diameter of 1.65 m and 10.7 m respectively, are inserted into each end of the barrel to extend coverage into forward regions. They generate a peak field 4.1 T. Both the barrel and end-cap toroids bend charged particles in the longitudinal plane.

2.3.3 Particle tracking

As mentioned in the previous section, magnets are fundamental to measure the momentum of charged particles. In fact, when a charged particle enters a magnetic field, it is deflected by the Lorentz force, leading to a curved trajectory, whose radius of curvature is proportional to the momentum of the particle. By measuring the curvature of the track, the momentum can be inferred.
The method used to measure the particle momentum is the sagitta method [36], where the sagitta s is defined as the distance between the midpoint of the chord connecting two points on the track and the track itself. The sagitta is related to the radius of curvature R, the length of the chord L and the angle θ under the arc by the following equation:

$$s = R\left[1 - cos\left(\frac{\theta}{2}\right)\right] \simeq \frac{L^2}{8R} \tag{2.15}$$

where the assumption of small angles was used in the final step.
Making use of the relation 2.2 between the radius of curvature R, the momentum p and the magnetic field B, the sagitta can be expressed as:

$$p = \frac{0.3BL^2}{8s} \tag{2.16}$$

As direct consequence of this latter equation, the reltive uncertainty on the momentum measurement is equal to the relative uncertainty on the sagitta measurement, $\frac{\Delta p}{p} = \frac{\Delta s}{s}$, which, using σ_x position uncertainty and N number of measured points, leads to:

$$\frac{\sigma_p}{p} = \frac{\sigma_x[m]}{0.3BL^2[Tm^2]}\sqrt{\frac{720}{N+5}}p[GeV] \tag{2.17}$$

From Eq. 2.17 it is evident that the momentum resolution can improve with a stronger magnetic field, a longer track length and a higher number of measured points but it worsens with higher momentum particles since $\frac{\sigma_p}{p} \propto p$.

2.3.4 The Inner Detector

The Inner Detector (ID) [37] is the closest system to the interaction point. Its function is to reconstruct the trajectories of charged particles, measure their transverse momentum and identify both primary and secondary interaction vertices of decays. Covering a pseudo-rapidity region of $|\eta| \leq 2.5$, 7.02 m long and 2.30 in diameter, it is inserted in the central solenoid (Sec. 2.3.2), providing the tracking sub-systems an external magnetic field of 2 T parallel to the beam axis. In order to balance both optimal resolution, material budget and radiation exposure, the ID utilizes three concentric silicon-based tracking technologies organized radially outward from the beam pipe: the Pixel Detector, the Semiconductor Tracker (SCT) and the Transition Radiation Tracker (TRT). In Fig. 2.12 a schematic of the ID is shown.

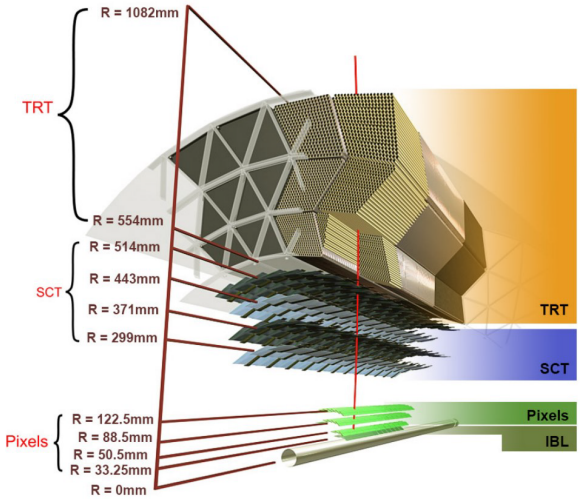


Figure 2.12 3D representation of the ATLAS inner detector. Figure form Ref. [38].

Pixel Detector

The closest system to the beam pipe is the Pixel Detector. It is composed of three pixel layers placed at distance of 50.5 mm, 88.5 mm and 122.5 mm from the IP respectively in the barrel. In the end-caps, there are installed three disks on each side located between $z = \pm 495$ mm and ± 650 mm. The pixel system has the finest granurality, delivering a spatial resolution of 10 μm in the (r, ϕ) plane and 115 μm along the longitudinal z-axis.
During LS1, an additional innermost layer was added at $r = 33.25$ mm. It is called the Insertable B-Layer (IBL) and it was added to improve the tracking performance, providing coverage up to $|\eta| = 2.58$ and to increase the radiation resistence of the ID.

SemiConductor Tracker

Outside the Pixel Detector, the SemiConductor Tracker (SCT) is positioned. It is composed of four barrel layers at distances between 299 mm and 514 mm and 18 overall end-cap disks. SCT delivers a spatial resolution of 16 μm in the (r, ϕ) plane

and 580 μm along the longitudinal z-axis.

Transition Radiation Tracker

The outermost region of the ID is composed of the Transition Radiation Tracker (TRT) situated between 554 mm and 1082 mm from the IP. Unlike the inner silicon-based detectors, the TRT consists of approximately 300,000 polyimide drift tubes, 4 mm each in diameter and equipped with a central gold-plated tungsten wire. The tubes are filled with a gas mixture based on Xenon-CO_2-O_2 (70%-27%-3%). TRT provides a single-hit spatial resolution of 130 μm in the barrel region with $|\eta| < 0.7$. In the end-caps, with $0.7 \leq |\eta| \leq 2.0$ the structure of the tubes is the one of disk, as for the other systems. The main feature of TRT is to provide continuous tracking by recording an average of 36 hits per charged track. Additionally, it is able to distinguish electrons from other objects by detecting the transition radiation photons which are emitted when charged particles cross the active materials inside the tubes.

2.3.5 Calorimeters

Calorimeters are used in ATLAS to measure the energy of particles by means of a destructive measurement on the particles. For this reason, calorimeters in ATLAS are placed outside the Inner Detector, allowing a complete reconstruction of the particle trajectory before it is absorbed [39].
Covering a region up to $|\eta| < 4.9$, the calorimetry system features sampling technology with dense absorbers alternated with active media to completely contain electromagnetic and hadronic showers. The idea behind this technique is to have a low probability of signals in the muon spectrometer from a particle different from a muon, which is the only particle that should cross the calorimeter without being absorbed (apart from not interacting particle such as neutrinos).
Liquid argon (LAr) serves as the active material, while being kept at a temperature of 88.5 K by three cryostats. When a charged particle crosses LAr, light is produced, which is collected by wavelength-shifting fibers and routed to photo-multipliers (PMTs) for the conversion to signal.
The calorimeter complex in ATLAS is divided into two components: the Electromagnetic Calorimeter (ECal) and the Hadronic Calorimeter (HCal). A schematic of the complex can be seen in Fig. 2.13.

Electromagnetic Calorimeter

The LAr ECal is composed of lead (Pb) absorbers (passive material) separated by LAr gaps (active material) with an accordion-shaped geometry. This particular layout is chosen to obtain high sampling rate while keeping a fast signal extraction time, ensuring uniform cover in ϕ. As previous systems, also ECal has a symmetric structure, with three main components: the barrel, with coverage $|\eta| \leq 1.475$ and the two end-caps in the range $1.35 \leq |\eta| \leq 3.2$. The region $1.37 \leq |\eta| \leq 1.52$ is known as *crack-region*, given the large amount of inactive material present in this area, which leads to a degraded energy resolution.

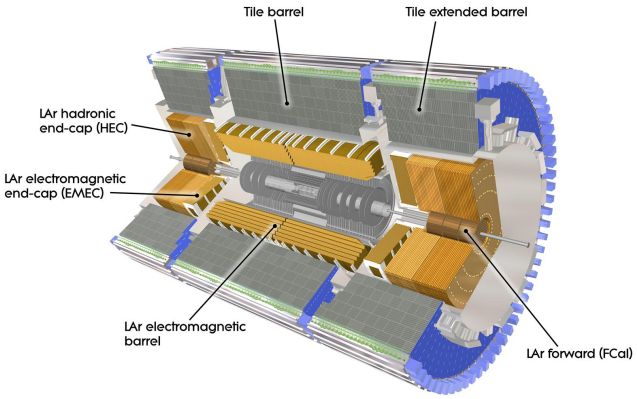


Figure 2.13 Schematic of the ATLAS calorimeter system. Figure form Ref. [35].

Based on the η region, the thickness of the ECal varies between 22 to 33 X_0, which is sufficent to contain the full electromagnetic showers.
The ECal is structured in three layers in the radial direction, each with a different granularity in η and ϕ. The first layer has a fine granularity in η, enabling the discrimination between showers coming from electrons, photons or neutral pions. The second layer collects the majority of the electromagnetic shower energy, with fine granulality in both coordinates. The third and final layer measures high-energy shower tails.
The ECal energy resolution has been measured during test-beams and, as a function of the beam energy, it is computed as:

$$\frac{\sigma_E}{E} = \frac{10.1\%}{\sqrt{E[\text{GeV}]}} \oplus 0.17\% \tag{2.18}$$

where the first term is the stochastic term, dominant at low energies, summed in quadrature with the term accounting for the non-uniform response of the detector.
A schematic representation of the ATLAS ECal is shown in Fig. 2.14.

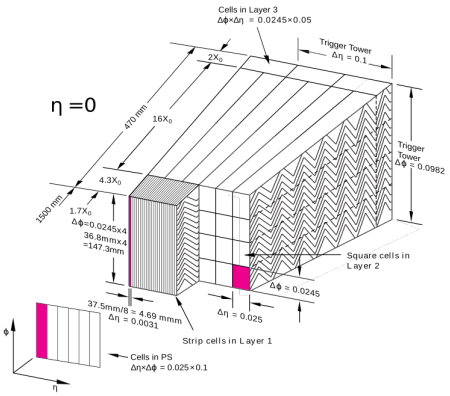


Figure 2.14 Schematic view of the accordion shape of the ATLAS ECal at $\eta = 0$. Figure form Ref. [40].

Hadronic Calorimeter

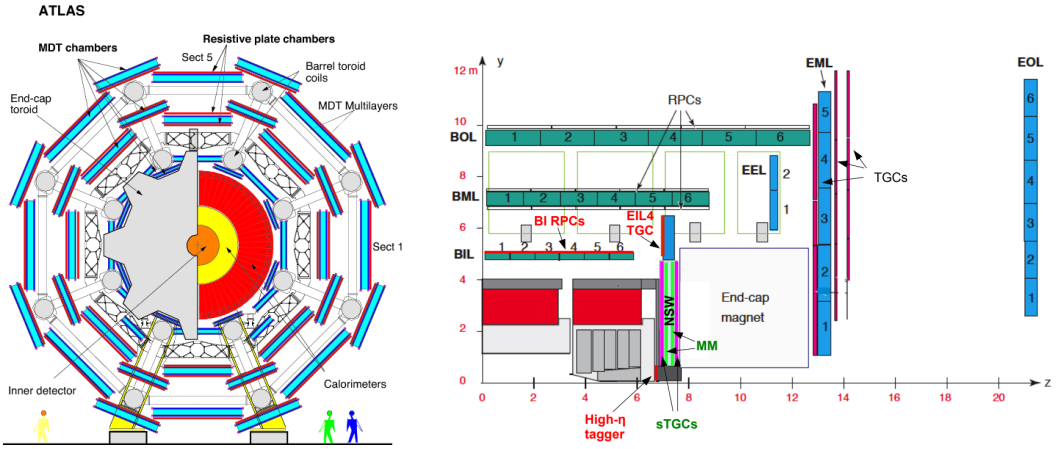
Hadronic energy measurements are handlead by the HCal, which is composed of three sub-sections: the Tile calorimeter, the Hadronic End-Cap Calorimeter (HEC) and the Forward Calorimeter (FCal). The Tile calorimeter is a sampling calorimeter with steel as absorber and scintillating tiles as active material, covering the central region of the detector with $|\eta| \leq 1.7$. With an inner diameter of 4.46 m and a outer one of 8.50 m, it covers 7.4 λ (the interaction length) at $\eta = 0$.
The HEC is based on two wheels per end-cap, with copper as absorber material covering up to 5.7 λ. With an outer diameter of 4.06 m, it covers the intermediate region of the detector with $1.5 \leq |\eta| \leq 3.2$.
In conclusion, the FCal covers the region $3.1 \leq |\eta| \leq 4.9$ and it is the closest component of the calorimeter system to the beam pipe. It is composed of three modules (FCal1, FCal2, FCal3), with copper as absorber for the FCal1, reaching 27.6 X_0, and tungsten for the other two, which allows to reach 90 X_0. The FCal is designed to resist to high radiation doses, being exposed to a high flux of particles, since a big fraction of collisions is associated with low momentum transfer.
The overall hadronic calorimeter resolution can be parametrized as:

$$\frac{\sigma_E}{E} = \frac{52\%}{\sqrt{E[\text{GeV}]}} \oplus \frac{1.6 \text{ GeV}}{E[\text{GeV}]} \oplus 3.0\% \tag{2.19}$$

where the first term, as for the ECal, is the stochastic term; the second one is associated to the noise from the read-out electronics and the last one is a constant term keeping tracks of unknown effects.

2.3.6 Muon spectrometer

The Muon Spectrometer (MS) forms the outermost layer of the ATLAS detector, with a total length of 44 m and an outer diameter of 25 m. Making use of the toroid magnets (Sec. 2.3.2) and the sagitta method (Sec. 2.3.3), it tracks particles with $p_T > 3$ GeV that are not absorbed by the different calorimeters in the region $|\eta| \leq 2.7$, which are mainly muons. Targeting a resolution of 10% for 1 TeV muon tracks, MS is required to measure sagittas down to approximately 500 μm, which is quite a challenge for the detector. As for the other ATLAS sub-detectors, the MS is organized in a barrel and in an end-cap region. The barrel covers the central region $|\eta| < 1.05$ and it consists of three concentric cylindrical shells placed at radii of 5 m, 7.5 m, and 10 m from the beam line. The two end-cap regions cover the forward zones, $1.05 < |\eta| < 2.7$ for tracking and $1.05 < |\eta| < 2.7$ for Level 1 trigger. The end-cap detectors are divided in Small Wheel, located in front of the end-cap toroids, and Big Wheel, behind them. They are mounted on large perpendicular wheels centered at longitudinal positions $|z| = 7.4$ m, 10.8 m and 21.5 m. A schematic view of the MS is shown in Fig. 2.15.
The tracking and trigger instrumentation follow a 16-fold azimuthal (ϕ) layout, matching the geometry of the toroidal magnet coils. Sub-detector stations are alternated large sectors, positioned between adjacent coils, and small sectors, covering regions directly aligned with the magnet structures. Precision tracking and



(a) (b)

Figure 2.15 Fig. 2.15a represents the transverse view of the ATLAS Muon Spectrometer. Fig. 2.15b shows the longitudinal (z, r) view of the MS, plane orthogonal to the magnetic field from the barrel coils and therefore banding plane for muons. Figure from [41].

momentum measurement rely primarily on Monitored Drift Tube (MDT) chambers. Fast triggering is taken care by Resistive Plate Chambers (RPCs).

Tracking chambers

Each MDT chamber contains aluminum drift tubes of 30 mm diameter and 400 μm thick wall, filled with an Ar:CO_2 gas mixture (93%:7%) at a pressure of 3 bar. A central tungsten-rhenium anode wire of 50 μm diameter is insterted in the tubes. It operates at 3080 V, providing a gas gain of order 10^4 and yielding a muon sagitta resolution of $\sigma_S = 45$ μm. The drift chambers are composed of multi-layers of MDTs, each one containing three or four tube layers separated by aluminum spacers. Optical alignment systems continuously monitor structural deformations to preserve the tracking resolution. Alignment represents the dominant source of uncertainty in high p_T muon measurements; with low momentum particles, the uncertainty is dominated by fluctuations in the energy deposit.
To sustain expected HL-LHC background conditions, the original Small Wheels were completely replaced for Run 3 by the New Small Wheel (NSW) system. It introduces two complementary thin-gap technologies across 16 sectors per wheel: resistive-strip MicroMegas (MMs) for precision tracking and Thin Gap Chambers (TGCs) for fast triggering. MMs modules employ a $\sim$ 130 μm amplification gap defined by a metallic micro-mesh, achieving a 100 μm spatial resolution.

Trigger chambers

In the barrel region ($|\eta| < 1.05$), doublets and triplets of Resistive Plate Chambers (RPCs) are mounted. They function as continuous gaseous detectors operating without wires, built from parallel bakelite plates separated by a 2 mm gas gap operating under an electric field of 4.9 kV/mm. RPC readout strips achieve an exceptional timing resolution of 1.5 ns and spatial resolution of 10 mm both in z

and ϕ direction.
In the forward end-caps ($1.05 < |\eta| < 2.4$), fast triggering is handled by TGCs, which are designed to work under high flux of particles. They utilize a 3.2 mm strip pitch to provide a 4 ns time resolution and a spatial resolution of 2-6 mm in the bending plane and 3-7 mm in the non-bending plane.

2.3.7 Trigger system

At the LHC, proton bunches collide every 25 ns, resulting in a collision rate of 40 MHz and a projected data occupancy of $60{\times}10^6$ MBs^{-1}. Clearly, the ATLAS detector cannot record all of these events. The ATLAS Trigger and Data Acquisition (TDAQ) system [41, 42] is designed to select interesting events for further analysis while discarding uninteresting ones, reducing the data rate by five orders of magnitude to a manageable level. This reduction relies on a two level scheme: Level-1 (L1) and High-Level Trigger (HLT). An overview of the ATLAS TDAQ system is shown in Fig. 2.16.

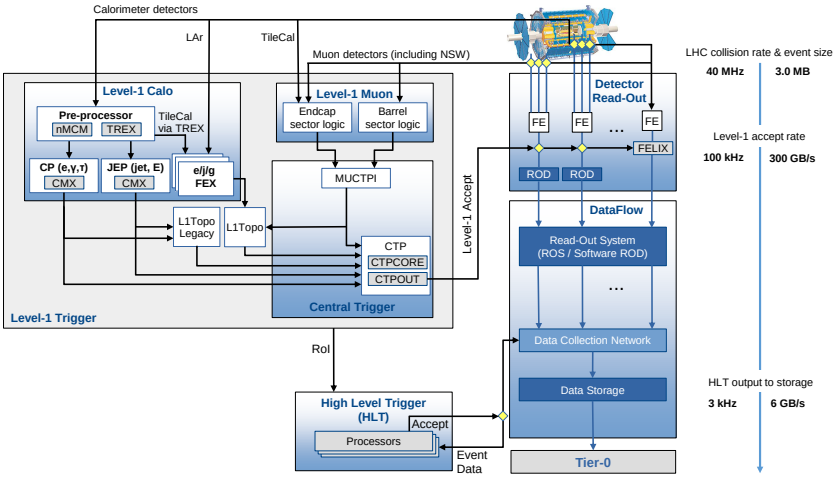


Figure 2.16 Schematic view of the ATLAS TDAQ system in Run 3. Figure form Ref. [41].

The L1 trigger, implemented using custom electronics, reduces the event rate to 100 kHz, which is the maximum rate that the ATLAS readout system can handle. It operates with a latency (time needed to take the decision) below 2.5 μs. The L1 Trigger searches for objects like electrons, photons, jets, τ-leptons, large E_T and large E_T^{miss} using reduced-granularity information from the calorimeters (L1Calo). Events containing high-p_T muons, instead, are hunted by muon system (L1Muon), composed of the RPCs and TGCs. If an event passes the L1 selection, it is sent to the HLT, together with some Regions of Interest (RoIs) identified by the L1 trigger. The HLT then processes the buffered events, operating at higher levels of detail than

the previous stage. It operates using full-granularity information from the ATLAS sub-detectors. To minimize network bandwidth, the algorithms initially focus on RoIs, defined in η and ϕ coordinates by the L1 stage. With a latency of 550 ms, the HLT reduces the event rate to approximately 3 kHz before storage.
Dedicated streams organize the stored events. While the primary physics stream retains full event information, the Trigger-Level Analysis (TLA) stream saves only partial reconstructed objects, bypassing bandwidth bottlenecks for high-rate processes. Accepted data is finally transferred to Tier-0 for offline processing.

2.3.8 The ATLAS Simulation Infrastructure

In order to interpret the data collected by the ATLAS detector and compare it with theoretical predictions, a detailed simulation of the detector response is required. The ATLAS simulation infrastructure is integrated within the Athena software framework [43], implementing the GEANT4 simulation toolkit to simulate the interaction of particles with matter [44].
This process is composed of two main steps: the *simulation* and the *digitization*. In the first one, Monte Carlo (MC) generators calculate the short-distance hard scattering, followed by parton showering, ignoring the detector geometry. After this, the generated particles with a proper lifetime $c\tau \geq 10$ mm are propagated through the detector geometry, using the GEANT4 toolkit. It simulates particles' interactions with the detector material and it records the energy deposits, together with the position and time of particles, referred as *hits*. Truth-level information is preserved throughout the process to track the exact parent-daughter decay trees.
The digitization step converts the previously cited hits into simulated electronic signals, emulating the readout electronics. To replicate realistic run conditions, the hits are combined with events due to cavern backgrounds and electronic noise. Soft QCD events are overlaid too, in order to model pile-up effects. The output of this stage are into Raw Data Objects, delivering a data format identical to the real detector readout. Consequently, identical reconstruction software used for real data are applied to simulated ones.
In conclusion, simulated events are re-weighted to match the average pile-up distribution observed in data.

Chapter 3

Hadron reconstruction in ATLAS

Before running the analysis, the signals recorded by the different ATLAS sub-detectors must be transformed into physical objects to deal with. In this chapter, the reconstruction of jets is presented, since they are the main objects of interest for the analysis conducted in this thesis.

3.1 Tracks reconstruction in ID

The ATLAS detector relies on a complex magnetic system, as described in Sec. 2.3.2, to bend the trajectories of charged particles, allowing for the measurement of their momentum. The starting point of these measurements is the reconstruction of charged particle trajectories, or *tracks*, in the Inner Detector (ID). They can be identified by groups of three-dimensional spatial points known as *hits*. Given the magnetic field configuration, tracks in the ID follow helicoidal trajectories, which can be described by the parameters: the transverse momentum p_T, the azimuthal angle ϕ, the polar angle θ, the impact parameter d_0, the longitudinal impact parameter z_0 and q/p, the momentum dividing the charge. Those parameters are graphically represented in Fig. 3.1.

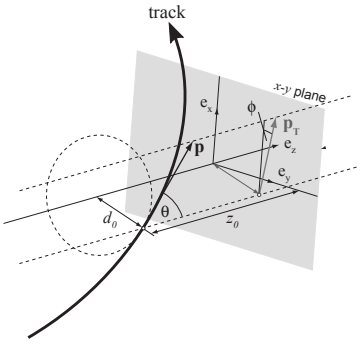


Figure 3.1 Parameters describing a track in the ATLAS detector. Figure form Ref. [45].

The tracks are reconstructed using an *inside-out tracking* strategy, starting from the innermost layers of the ID and moving outwards, with the requirement of at least 8 hits and $p_{\mathrm{T}} > 500$ MeV . The reconstruction algorithm begins by identifying clusters of hits in both the Pixel and SCT sub-detectors. Those clusters then form *track seeds*, which are composed by triplets of them. Using a combinatorial Kalman filter, the track seeds are extended into full track candidates, on which a global χ^2 minimization fit is performed to determine the initial trajectory parameters. After adding compatible hits from the TRT sub-detector, if the requirements are satisfied, another fit with higher precision is performed to refine the track parameters.
The tracks which have been reconstructed as described above, are then stored and used to reconstruct the primary interaction vertex (PV). Among all vertices, the one with the largest scalar sum of squared track transverse momenta, $\sum p_{\mathrm{T}}^2$, is identified as the vertex of the hard scattering (HS) and its position is taken as reference point to reconstruct all the other objects.

3.2 Jets reconstruction

As described in Sec. 2.2, the scattering of two partons in a *pp* collision produces high-energy quarks and gluons, which undergo a process of parton showering and hadronization, producing collimated sprays of particles, which are known as *jets*. A schematic of the jets nature is presented in Fig. 3.2.

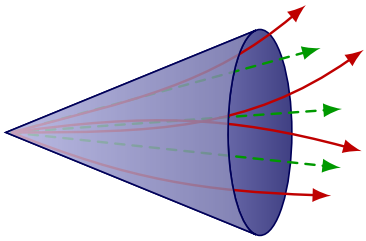


Figure 3.2 Drawing of a jet, showing the cone of particles produced by the hadronization of a parton.

Jets costituents then form hadrons due to the QCD colour confinement, which are detected by the ATLAS sub-detectors.
The reconstruction of jets is a complex process, which requires the combination of information from different sub-detectors. The diffent components of the spray are detected and reconstructed as different objects, which are then combined to form the final jet object. The combination is the not trivial part and dedicated algorithms have been developed to perform it.

3.2.1 Jet constituents reconstruction

The reconstruction of jets starts with the reconstruction of their constituents, which are the physical objects detected by the ATLAS sub-detectors. This process uses different algorithms depending on the target physics regime and kinematic regime,

combining the measurements of the ID with those from calorimeters. The common starting point is the clustering of calorimeter cells, which are grouped into *topological clusters*.

Topoclusters

Topoclusters, or jet topological clusters, are three-dimensional clusters of energy deposits in calorimeter cells. They dynamically adapt their spatial volume based on the topology of the energy shower across both the LAr and Tile calorimeter layers. The formation of topoclusters is controlled by the parameter ζ_{cell}, the cell signal significance[46], computed as the ratio between the signal E_{cell} and the average noise $\sigma_{noise,cell}$ in the cell:

$$\zeta_{cell} = \frac{E_{cell}}{\sigma_{noise,cell}} \tag{3.1}$$

All the cells with $\zeta_{cell} \geq 4$ are considered as seeds for the topocluster formation. Starting from these seed cells, the cluster is grown by adding adjacent cells with lower significance, $\zeta_{cell} \geq 2$. Finally, all the cells with $\zeta_{cell} \geq 0$ are added to the cluster if they are adjacent to a cell already added to the cluster.
In the case of overlapping particle showers, the topocluster can contain multiple local energy maxima, which are then split into two separate clusters, with the requirement $E_{\text{cell}} > 500$ MeV.
To each topocluster is assigned a four-momentum P_{clus} and it is treated as a pseudo-particle. In fact the mass is set to zero, since there is no information about the mass of the particles producing the energy deposits, and the energy is computed as the sum of the energies of the cells in the cluster:

$$E_{\text{clus}} = \sum_i E_{\text{cell},i} \tag{3.2}$$

The direction parameters of the cluster, η_{clus}, ϕ_{clus}, are determined from the energy-weighted barycenter of the cluster relative to the nominal origin of the ATLAS coordinate system:

$$\eta_{\text{clus}} = \frac{\sum_i |E_{\text{cell},i}|\eta_{\text{cell},i}}{\sum_i |E_{\text{cell},i}|}, \qquad \phi_{\text{clus}} = \frac{\sum_i |E_{\text{cell},i}|\phi_{\text{cell},i}}{\sum_i |E_{\text{cell},i}|} \tag{3.3}$$

Absolute cell energies are utilized in the denominator weighting to prevent directional bias from negative cell energy fluctuations induced by noise or pile-up subtraction. Those kinematic variables are used to distinguish between electromagnetic and hadronic showers. In fact, as described in Sec. 2.2.3, hadronic showers contain also EM components, resulting in the fact that the reconstructed clusters are only partially hadronic. For this reason, after Local Cell Weighting (LCW) calibrations of the topoclusters, a probability $P_{\text{EM}}^{\text{clus}}$ is assigned. It tells us the probabilty that the deposit is electromagnetic versus hadronic. Topoclusters with $P_{\text{EM}}^{\text{clus}} < 0.5$ are identified as hadronic, otherwise they are identified as electromagnetic.

Particle Flow (PFlow)

The Particle Flow (PFlow) algorithm [47] is built to optimize the reconstruction of jets with lower transverse momentum, 0.5 GeV $\leq p_T \leq$ 40 GeV. This algorithm relies on the fact that at lower energies, the momentum resolution is higher than the energy one. This feature is evident in the formulas associated with the two resolutions, Eq. 2.17 and Eq. 2.19, which show the different scalings with respect to the momentum ($\sigma_{p_T}/p_T \propto p_T$) and energy ($\sigma_E/E \propto 1/\sqrt{E}$) respectively.
The algorithm begins by the selection of the tracks to target: 0.5 GeV $\leq p_T \leq$ 40 GeV, $|\eta| < 2.5$, at least 9 silicon hits, zero pixel holes and $E_{clus}/p_{trk} > 0.1$ (E_{clus} is the energy of the matched topocluster and p_{trk} is the momentum of the track). After this, PFlow subtracts from the matched topoclusters the expected energy deposit from charged hadrons of the selected tracks, replacing it with the expected energy deposit estimated from the track momentum p_{trk}, ensuring higher resolution. If the residual energy of a topocluster after subtraction is consistent with fluctuations expected from a single-particle signal, the remnant cluster is completely removed. If significant energy remains, it is treated as a neutral energy remnant.
The finalized PFlow constituents consist of selected charged tracks for charged hadrons and un-subtracted or remnant topoclusters for neutral hadrons and photons.

Track Calo-Clusters (TCCs)

The angular separation between the products of hadronic decays is highly reduced in the case of highly boosted regimes. The high track density and calorimeter energy overlap degrade the performance of the PFlow algorithm. Track-CaloClusters (TCCs) offer a solution to this limitation [48]. They combine the track directional precision with topocluster energy measurements. TCCs match the tracks, which are extrapolated from the primary vertex to calorimeters, with topoclusters if their relative angular separation satisfies:

$$\Delta R < \sqrt{\sigma^2_{\text{clus}} + \sigma^2_{\text{trk}}} \tag{3.4}$$

In the equation above σ_{clus} is the topocluster width and σ_{trk} represents the track extrapolation uncertainty. Unlike PFlow, TCCs do not perform energy subtraction; instead PU is removed.
The TCC objects preserve fine angular substructure within large-radius ($R = 1.0$) jets, improving jet mass and substructure observable resolutions (τ_{21}, D_2).

Unified Flow Objects (UFO)

While PFlow excels at low p_T and TCCs dominate in highly boosted regimes, Unified Flow Objects (UFO) synthesize the two methodologies into a single unified input [49].
UFO are built by first executing the PFlow algorithm, where charged PFlow objects associated with pile-up interaction vertices are removed. From the remaining neutral PFlow topoclusters uncorrelated soft radiation is removed. The surviving neutral clusters and hard-scatter ID tracks that were not subtracted during the initial step are passed into the TCC cluster-splitting algorithm.

By combining PFlow subtraction at low p_T with TCC angular splitting at high p_T, UFO inputs optimize small-radius ($R = 0.4$) and large-radius ($R = 1.0$) jet performance simultaneously across the full momentum spectrum.

3.2.2 Constituents combination and jets formation

After having reconstructed the constituents of the jets, they are combined to form the final jet object. The combination is performed applying sequential recombination algorithms, which must satisfy two main requirements: *infrared* and *collinear safety*. The first one, infrared safety, requires that the addition of a soft particle to the jet cluster does not impact on the jet itself. The second one, collinear safety, requires jet invariance under the splitting of a constituent particle into two collinear particles. The sequential recombination algorithms are based on the iterative combination of pairs of constituents (i, j), which are combined into a single object if their distance d_{ij} is smaller than the distance d_{iB} between each of them and the beam. These distances are defined as:

$$d_{ij} = min(p_{Ti}^{2k}, p_{Tj}^{2k}) \frac{\Delta R_{ij}^2}{R^2} \ , \quad \mathrm{d_{iB}} = \mathrm{p_{ti}^{2k}} \tag{3.5}$$

where R is the radius parameter of a jet and p is an additional parameter about the power of energy. p could be equal to 1, falling in the k_t algorithm, 0 for the Cambridge/Aachen and -1 in case of anti-k_t algorithm [50].
For the purpose of this thesis, the anti-k_t algorithm is used, since it produces regular cone-like jets around hard seeds and it is less sensitive to soft radiation. Typically in ATLAS large-radius jets are reconstructed with a radius parameter of $R = 1.0$ to collect energy deposits from boosted decays. Small-radius jets, instead, are built with $R = 0.4$ and are usually used to identify highly energetic partons generating a shower.
In Fig. 3.3 an example of anti-k_t application with large-radius jet $R = 1.0$ is shown in the case of a parton-level event associated with soft radiation.

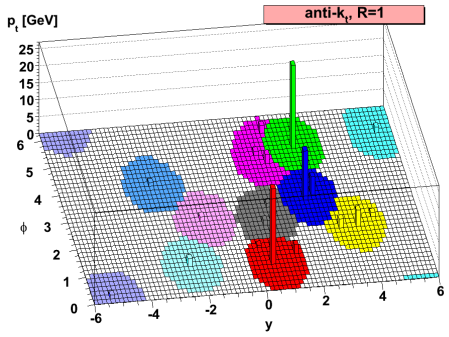


Figure 3.3 Example of anti-k_t application with large-radius jets. Figure from Ref. [50]

3.2.3 Jet calibration

To ensure accurate measurements, a structured calibration workflow is implemented on the reconstructed jets. In the first place, a correction due to the presence of

pile-up is applied. It consists in the subtraction to the jet energy, based on the average energy contribution from PU (ρ) and the jet area. This correction is applied making use of MC simulations.
The second step is the Jet Energy Scale (JES) calibration, which accounts for detector response and resolution effects, as well as for the presence of inactive material in the calorimeter. JES is applied matching the reconstructed jets with the corresponding truth jets in MC simulations, and it is derived as a function of the jet p_T and η.
Another calibration, known as Global Sequential Calibration (GSC), is applied via sequential multiplicative corrections, to absorb residual differences in the ATLAS response due to variations in the energy distributions of constituents or the jet flavor and topology. GSC is derived from energy measurements in certain sections of the experimental apparatus.
Finally, an *in-situ* calibration is applied to correct for residual differences between data and MC simulation, using well-measured reference objects such as photons or Z bosons.
Associated to each approach cited above, a systematic uncertainty is derived, which is then propagated to the final analysis. At low p_T the dominant uncertainty is due to PU and different detector response; JES, which is below 5%, is the main source of uncertainty at high momentum ($p_T > 400$ GeV).
The JER, jet energy resolution, can be espressed, as a function of p_T in a form similar to those for the energy resolution of the calorimeters:

$$\frac{\sigma_{p_T}}{p_T} = \frac{S}{\sqrt{p_T}} \oplus \frac{N}{p_T} \oplus C \tag{3.6}$$

with S the stochastic term associated with detector response, N the noise term and C the constant term associated with jet formation, which is the major contribution at high momentum.

3.3 Jet substructure variables

A single hadronic final state can be reconstructed in different ways, from multiple small-radius jets to a single large-radius jet, depending on the boost regime of the event. In the latter case, the reconstructed jet contains the highly collimated decay products of the original parton, which form a subtructure within the jet, differentiating it from the case when the jet is produced by a single parton. In Fig. 3.4 an example is shown of the top quark decaying hadronically in the channel $t \to Wb \to q\bar{q}'b$, in three different boost regimes. In particular, from Fig. 3.4c it is visible that the highly boosted red large-radius jet contains a subtructure of three smaller jets, differentiating it from the case of a single parton small-radius jets, as shown in Fig. 3.4a.
Subtructure can then discriminate between jets originating from different physical processes, such as the decay of the top quark in three "prongs"[1], the decay of a W/Z boson in two "prongs" or a single-prong QCD or BSM jet. Specific variables, based on the kinematic properties of the jet constituents and their spatial distribution within the jet, have been developed to allow this classification. Those variables

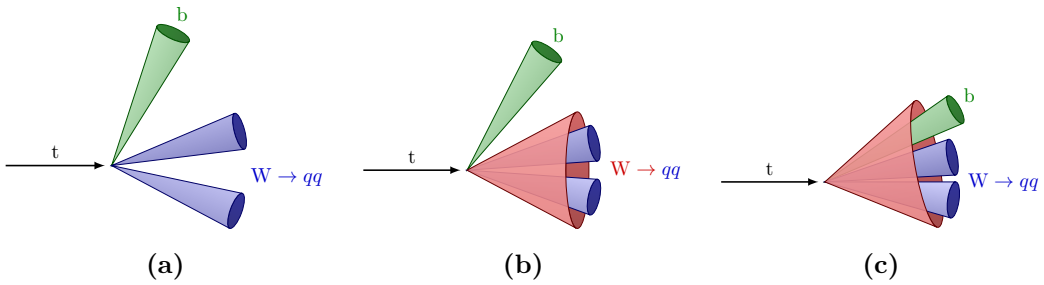


Figure 3.4 Diagram illustrating different boosted jet regimes of the hadronic top decay and their reconstruction. The three subfigures represent (a) a lowly boosted top quark, (b) a moderately boosted top quark, and (c) a highly boosted top quark. In each case, the decay products of the top quark are shown, along with the corresponding jet reconstruction.

presented below are used in the analysis conducted in this thesis, as comparison with the results obtained from the Anomaly Detection approach.

3.3.1 N-Subjettiness (τ_N)

The N-subjettiness [51], denoted as τ_N, is used to quantify the compatibility of a jet reconstruction with an N-prong hypothesis. Following the jet reconstruction as described in the previous section, τ_N is calculated as:

$$\tau_N = \frac{1}{R_0 \sum_i p_{\mathrm{T},i}} \sum_i p_{\mathrm{T},i} \min\left\{\Delta R_{1,i}, \Delta R_{2,i}, \ldots, \Delta R_{N,i}\right\} \tag{3.7}$$

where i runs over all jet constituents, R_0 is the jet radius and $\Delta R_{k,i}$ represents the distance between constituent i and subjet axis k. Small values of τ_N indicate that the radiation is aligned with the direction of the N subjects, therefore the jet is compatible with an N-prong hypothesis, while large values of τ_N indicate that the jet are compatible with the $N+1$-prong and higher hypothesis. The ratio of two subjettiness is usually preferred to discriminate between different prong hypotheses. It is defined as:

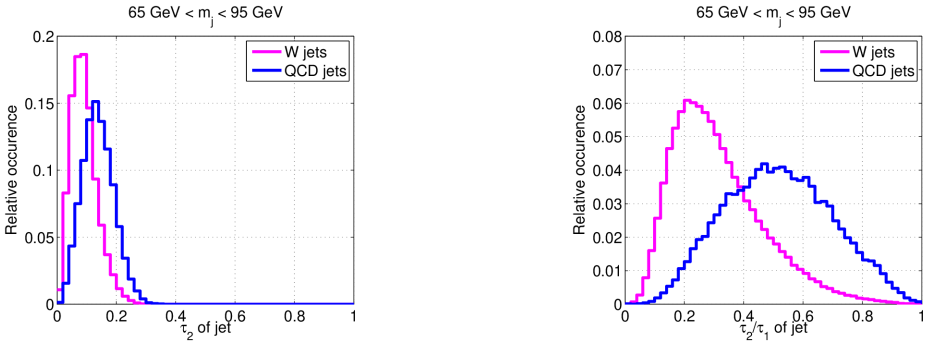
$$\tau_{N,N-1} = \frac{\tau_N}{\tau_{N-1}} \tag{3.8}$$

The ratio $\tau_{21} = \tau_2/\tau_1$ can isolate two-prong vector boson decays such as W/Z from single-prong QCD jets. $\tau_{32} = \tau_3/\tau_2$, instead, can be used to tag three-prong top quark decays.
In Fig. 3.5 example distributions of τ_2 and τ_{21} are shown for jets produced in W and QCD processes. It shows that the two-prong hypothesis is more compatible with the former than the latter and this can be identified more easily using the ratio τ_{21}.

3.3.2 Energy Correlation Functions (ECFs) and D_2

Energy Correlation Functions (ECFs) [52, 53] are using, once again, to tag jets with substructure without necessitating the definition of candidate subjet axis. Referred

[1] *Prong* is the colloquial term used to refer to the constituents of the jet substructure.



(a) **(b)**

Figure 3.5 (a) Distribution of τ_2 for jets produced by W and QCD. (b) Distribution of τ_{21} for the same entities. Figure from Ref. [51]

to the jet J, they are defined as:

$$\mathrm{ECF}_0 = 1, \quad \mathrm{ECF}_1 = \sum_{i \in J} p_{\mathrm{T},i} \tag{3.9}$$

$$\mathrm{ECF}_2(\beta) = \sum_{i<j \in J} p_{\mathrm{T},i} p_{\mathrm{T},j} (\Delta R_{ij})^{\beta} \tag{3.10}$$

$$\mathrm{ECF}_3(\beta) = \sum_{i<j<k \in J} p_{\mathrm{T},i} p_{\mathrm{T},j} p_{\mathrm{T},k} (\Delta R_{ij} \Delta R_{ik} \Delta R_{jk})^{\beta} \tag{3.11}$$

where β, usually set to 1, is an angular weighting exponent. A combination of these function is used to define D_2, which is a powerful discriminant to identify two-prong substructure within jets. The D_2 variable is derived as:

$$D_2 = \frac{\mathrm{ECF}_3 \cdot (\mathrm{ECF}_1)^3}{(\mathrm{ECF}_2)^3} \tag{3.12}$$

Similarly to τ_2, lower values of D_2 are associated with a higher probability of the jet being produced by a two-prong decay, while higher values indicate more probable single-prong origin. An example of the D_2 distributions for jets produced by Z and QCD processes is shown in Fig. 3.6.

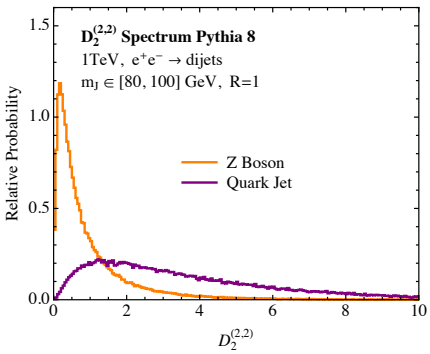


Figure 3.6 Distribution of D_2 for jets produced by Z and QCD. Figure from Ref. [53]

Chapter 4

Anomaly Detection

As previously discussed in Sec. 1.2, SM is not a complete theory and New Physics beyond it should exist. The usual model-dependent searches for BSM physics rely on MC simulations generated for specific BSM hypotheses to define signal-enriched regions. They provide optimized sensitivity for the specific process although they remain blind to a broader scenario and they have not yet produced any significant discoveries. This motivates the exploration of alternative approaches.
Model-independent searches are the most common solution, as they do not rely on specific theoretical assumptions [54]. They have been used for a long time among the High Energy Physics (HEP) community to derive limits on the production cross-section of BSM signals. Despite their name, these searches were still too dependent on the theoretical assumptions or on the simulation for the event selection and the background modelling, limiting their power. Beyond that, their sensitivity was limited by the look-elsewhere effect [55].
Anomaly Detection (AD) addresses these constraints by identifying rare patterns without the assumption of a explicit signal hypotheses.

4.1 What is Anomaly detection?

Anomaly Detection (AD) [56, 57] is a term used to refer to a series of algorithms involved in the identification of patterns or features among data deviating significantly from a baseline reference distribution. Such deviations are referred to as *anomalies*, *outliers* or *signals*. The baseline, instead, is addressed as *normal* or *background* data. Given their behiavour, anomalies could be generated by a distinct mechanism with respect to the one producing the background set and for this reason it is worth to investigate them.
AD relies on the assumption that anomalies are rare and they represent a small portion of a whole dataset. This assumption is realistic because it is what happens everyday in airports, finance, healthcare: normal events are way more common then deviations.
In order to compensate the loss of sensitivity due to their model agnostic nature and to maximize the potential to identify anomalies, AD algorithms need to learn and recognize huge amounts of normal data. By doing so, they should be able to flag more easly an event as anomalous if it deviates from the learned normal



Figure 4.1 Explicative drawing of anomalies. Figure adapted from Ref. [57].

behaviour. Given the complexity and the high dimensionality of the task, given the quick development in the field, Machine Learning (ML) and Deep Learning (DL) techniques are the most suitable tools to implement AD tasks. These are, in fact, able to learn from past by using vast datasets.

4.2 Deep Learning

Deep Learning (DL) tries to replicate the behaviour of a human brain and, analogously, its basic unit is named *neuron* [58, 59]. From its dendrites it receives input values; they are processed and then summed in a weighted sum. After applying an activation function, the signal is sent in output through the axon, towards the next dendrites. Neurons are organized in layers, constituting a network. The first one is defined *input layer* and it receives original data as input; the last one is the *output layer* and its output is the output of the model. Between them there are hidden layers, performing mathematical computations. A model with just one hidden layer is Perceptron. If the number of layers is large, the model is defined as Deep Neural Network (DNN). In Fig. 4.2 a graphical representation of a Perceptron and a DNN is shown.

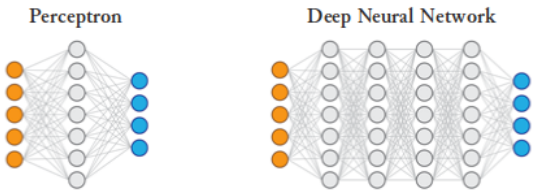


Figure 4.2 Comparison between a Perceptron and a Deep Network. Figure from Ref. [58]

4.2.1 Mathematical definition

The aim of a model training is to find the best function f that connects input $\mathcal{X}$ and output data $\mathcal{X}$, $f : \mathcal{X} \to \mathcal{Y}$. In order to achieve this goal, the function connecting a

layer to the following is defined as [60]:

$$O_i^L = \sigma\Big(\sum_j W_{ij}^L I_j^L + B_i^L\Big) \tag{4.1}$$

where I_i^L is the input vector, O_i^L the input vector, W_{ij}^L is the weight matrix (which has to be optimized during training), B_i^L represents the bias vector for the layer L and σ is the activation function applied before signal leaves the neuron. When dealing with an hidden layer, the assumption $I_i^L = O_j^{L-1}$ is made. The activation function is a mathematical function setting a threshold to determine whether the neuron is active or not. Among the most used ones and of interest for the purpose of this study, there is the Parametric Rectified Linear Unit (PReLU) [61]. It is computed as follows:

$$\mathrm{PReLU}(x) = \begin{cases} x, & \text{if } x \geq 0 \\ ax, & \text{otherwise} \end{cases} \tag{4.2}$$

where a is a learnable parameter.

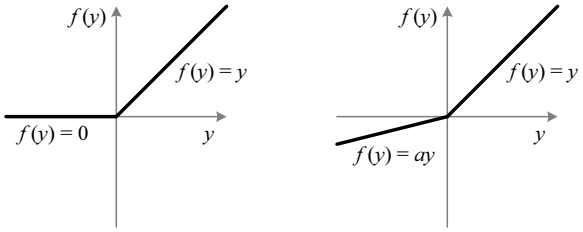


Figure 4.3 Comparison between ReLU (Rectified Linear Unit) on the left and PReLU on the right. Figure from Ref. [61]

4.2.2 Loss function

The model implementation is based on math and also the process of it learning does. A loss function $\mathcal{L}$ is defined from the distance between the input and the output of the model. During the training phase (one of the steps of the learning), the goal is to minimize this function adjusting weights of the model. A possible method to do so is the stochastic gradient descent (SGD). Defining $\mathbf{p} \in \mathbb{R}^s$ as the vector containing weights and biases, SGD is computed as [58]:

$$\mathbf{p} \to \mathbf{p} - \eta \left(\frac{\partial \mathcal{L}}{\partial p_r} \Delta p_r \right)_{x^{\{i\}}} \tag{4.3}$$

where η is the learning rate, an hyperparameter that has to be optimized, computed as the ratio between the step size and the absolute value of the gradient. Δp_r is a perturbation associated to the step. To calculate the gradient back-propagation is applied: weights are computed while forward feeding the model and then computed again when data passes backwards. Values are combined to present the best estimation of weights.

4.2.3 Learning paradigms

ML and DL methods can be classified based on how the labels of data are used to train models. Main categories are: *Supervised, Unsupervised, Semi-Supervised and Weakly-Supervised Learning.*[57]
In Supervised Learning, shown in figure 4.4, labeled data is used to train DL models and it can be implemented only with sufficient normal and anomalous data.



Figure 4.4 Data used for Supervised Learning. Figure from Ref. [57]

Unsupervised Learning is based on not labeled data, as shown in figure 4.5, and models implementing it are trained to find a structure among the input features. Then model performance is tested using a mixed dataset containing both background and signal data.
Semi-Supervised Learning is a hybrid of previous techniques, implemented in the most varied ways.
Weakly-Supervised Learning, in conclusion, is associated with trainings executed using noisy samples.



Figure 4.5 Data used for Unsupervised Learning. Figure from Ref. [57]

4.2.4 The steps for a model to learn

The process of training a DL model is divided into three main phases: training, validation and test loop. They are defined "loop" because in each of them a different section of the dataset is used and the code loops over it.
The most important stage is the training loop. In this phase, parameters of the model are adjusted in order to best fit to data. By doing so, DL model should be able to recognise very well normal data and their structure.
The second step is the validation loop, where a valuation of the performance of the model is elaborated using the validation set. The valuation and the search for the best weights of the model are executed computing a given metric, such as the distance between output and input over training and validation set. When this distance is minimum within the validation loop, the model's weights are saved.
The trained and tuned model is then tested, freezing the model's weights and using a dataset containing unseen data, the test set. The purpose of this phase is to validate results previously obtained and get a preview of what will happen once the model will be running with completely new data.

4.2.5 Estimating the model's performance

In order to understand how well a model is performing, it is necessary to define a quantitive estimator. Since an AD task is generally a classification task (normal data or not), the most common estimators are the Area Under the Curve (AUC) [62] and the Receiver Operating Characteristic (ROC) curve [63] They are computed starting from the confusion matrix, where four groups are presented to quantify misclassification errors: true positive (TP), true negative (TN), false positive (FP) and false negative (FN). From them, rates are calculated:

$$TPR = \frac{TP}{TP + FN} \ , \quad FPR = \frac{FP}{TN+FP} \tag{4.4}$$

where TPR is the True Positive Rate and FPR is the False Positive Rate. ROC curve, as shown in figure 4.6, is a plot showing the trade-off between TPR and FPR at various classification thresholds. The curve is usually over the line where TPR=FPR and the closer it is to the top left corner, the better the model is learning. AUC is, as the name suggests, the area under the ROC curve; it is computed using an integration method, for instance the trapezoidal rule. A random classifier can only obtain a AUC=0.5 (TPR=FPR); a perfect classifier flies to a $AUC \to 1$.

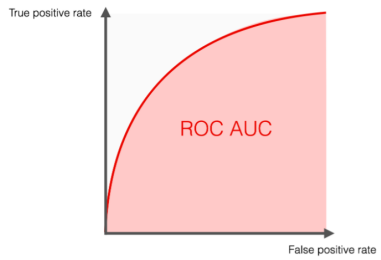


Figure 4.6 ROC curve and AUC explained

4.3 Anomaly Detection in High Energy Physics

Searches for NP could gain a reasonable benefit from the implementation of AD techniques, given the multiple limitations of the field. There are two main categories by which AD be used in this context: *outlier detection* and *overdensity estimation* [64]. The first one is based on the identification of events that are outliers, deviating for the common behaviour of the dataset, usually in the tails of distributions. This region is typically challenging to model and it is often dominated by statistical fluctuations, making it difficult to distinguish between real anomalies and random noise. The second one, instead, is based on the identification of a localised excess of data points over an expected smooth background.
The increasing popularity of AD in HEP is due to its model agnostic nature, which allows to scan a bigger range of events, without any stringent assumption on the theoretical model, just hunting from outliers. Together with this, large unlabeled datasets have been collected by collider experiments, boosting the usage

of unsupervised learning techniques.
In this approach, the data is divided into two categories: *background*, containing Standard Model events, and *signal*, including the hypothetical NP.
Typically, before the raise of AD, the searches for NP were conducted on simulated MC data in the first stage, applying a proper event selection to define a region of interest, called Signal Region (SR), in the phase space of parameters where the signal is expected to be found. Then, after validating the found parameters, the search was performed on real data, looking for an excess of events in the SR with respect to the background prediction, as shown in figure 4.7.
This approach relied strongly on simulated MC data.

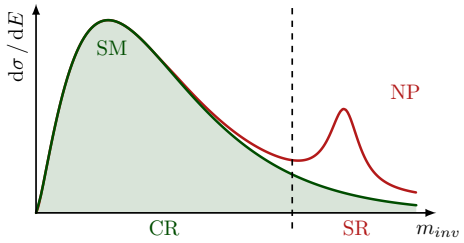


Figure 4.7 Schematic drawing of a bump hunt search, where the SR and the Control Region (SR) are defined to isolate a potential signal from the background.

With the implementation of AD, the dependance on MC simulations can be reduced. The DNN, in fact, can be trained and validated directly on real data, in a CR where the presence of signal is not expected. For this reason, similar implementations of AD are referred to as *data-driven* approaches.
Once the DNN is trained, it is used to compute the Anomaly Score (SR) of the data, a distirbution of the probability for it to be anomalous, as shown in Fig. 4.8. AS is defined as a distance between the input and the output of the model. The assumption to be verified is that anomalous data will have a different distance with respect to the background, making the two AS distributions separated: greater the separation is, easier it will be to distinguish signal events from background ones and discarding the latter.

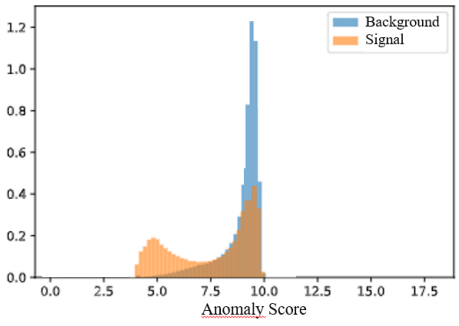


Figure 4.8 Example of AS distribution plot

On the AS distributions Working Points (WPs) are defined. They are used to reject

a given amount of background: by selecting events with score greater (or lower, depending on the definition of AS) than a fixed threshold, the number of background events is reduced, hoping to have enhanced the signal. After the WPs computation, the second part of the search for WP can begin, with the application of common statistical analysis tools to derive an estimatore of the impact tof AD on banchmark signals.

4.4 Towards the statistical analysis

To better understand Anomaly Detection and how it is used to execute hunts for BSM processes, we can use an example.
Assuming we need to find a church in a town we do not know, without any technology. Instead of desperately walking around, we decide to climb an hill where we know that we can scan the whole town without so much effort. Once on top of the hill, it is possible to identify the neighbourhood of the church or the "spatial region" where it is situated. We can also get some information we would not have got in the town, such as the colour of the roof or of the facade of the building we are looking for. Then we climb down the hill and go straight to the point we observed. Thanks to the details we obtained up there, we are able to find the church quite quickly and more easily.
In other words, the church represents New Physics to be found, climbing the hill is training a DNN within the Anomaly Detection approach, climbing down is using statistical data analysis techniques. Such approach has a huge potential.



Figure 4.9 View of Rome from the Janiculan Hill. From it, it is possible to scan the whole city and identify the neighbourhood of the Basilica of Saint Mary Major, for instance.

To apply Anomaly Detection it is necessary to hypothesize that anomalies are rare and they represent a small portion of a whole dataset. This assumption is realistic

in everyday life, where normal events are way more common then deviations, but also in Particle Physics: if anomalies were more frequent, we would have already observed NP through common data analysis.

Chapter 5

ATLAS analysis with AD in hadronic final states

The final and crucial chapter of this thesis is dedicated to present the implementation of the Anomaly Detection approach within an ATLAS analysis. This study searches for boson resonances associated with BSM physics decaying into fully hadronic final states in the described detector, making use of the AD technique.
The goal of this analysis is to prove the feasibility of the AD approach in a full analysis, exploring its generalization capabilities, while hunting for New Physics and trying to improve the sensitivity of the search with respect to the state-of-the-art methods.
At the moment, the status of the analysis is the following. The R&D phase on the MC sample for the DNN model can be defined as concluded, as it is for the statistical analysis and modelling on the MC dataset. The preparation of the field to move towards the usage of the real data is currently on going, expecting the switch in dataset to happen soon.
In the following sections, an overiew of the analysis is presented, with a description of the dataset and the Graph Neural Network model used for the AD implementation. After this, an important portion of the chapter is dedicated to the description of the original work related to this thesis, which is the execution of the statistical analysis on the MC dataset, with the aim of testing the AD approach in a full analysis.

5.1 The dataset

The analysis focuses on exploring the data collected by the ATLAS detector during the LHC Run-3 from 2022 to 2024. It corresponds to a total integrated luminosity of 165 fb^{-1}, with a center-of-mass energy of $\sqrt{s} = 13.6$ TeV [29].
Before running the full analysis pipeline on the real data, a preliminary study is performed on a MC dataset, which is used to train and validate the AD model and to test the statistical tools developed. The latter aspect is the goal of this thesis, which aims to conduct the analysis on the MC sample and validate the tools. In this preliminary stage, a sample simluating 2023 events is used, the one developed during *MC23d* campaign.

The composition of the dataset is described in the following.

5.1.1 Final states

This analysis hunts for BSM physics in the form of heavy resonances with masses in the TeV range which decay into quarks forming final states with two large-radius jets ($R = 1.0$). Such signatures are chosen because they are predicted by several NP models and they are characterized by a high Branching Ratio (BR), from which the analysis sensitivity could benefit. Together with this, the high mass of the resonances results in highly boosted decay products, which are collimated and reconstructed as single large-radius jets each. This leads to a final state where jets can present a complex substructure, which can be exploited by the AD model to identify possible anomalies and NP candidates.

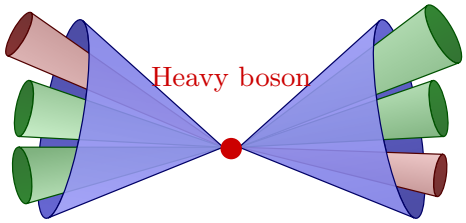


Figure 5.1 Schematic drawing of a heavy boson decaying into two large-radius jets (blue cones) with substructure, both normal (green cones) and anomalous (red cones).

5.1.2 Signal processes

Despite the model-agnostic nature of the AD approach, a set of benchmark signal processes has to be chosen to test the sensitivity of the technique. In this analysis, two main physical processes are considered: Heavy Vector Triplet (Sec. 1.2.2) and Dark Jets (Sec. 1.2.3).
The Dark sector is probed using a sample of a Z' boson with mass $m_{Z'} = 3.5$ TeV, decaying into a pair of dark quarks with a mass $m_{q_D} = 20$ GeV each .
HVT, instead, are explored with three different processes. The first one is the decay of a heavy vector boson V, assumed to be the candidate Y boson, into a BSM vector boson X and the SM Higgs boson H. The masses of the V and X bosons are varied in order to explore different kinematic configurations, among $m_V = [2.3, 3.4, 5.0, 6.0]$ TeV and $m_X = [100, 200, 2500, 400]$ GeV respectively. The mass of the Higgs boson, instead, is fixed to its SM value of $m_H = 125$ GeV.
The second HVT process considered is the decay of a heavy vector boson V, into a pair of SM W bosons. Once again, the mass of the V boson is varied among $m_V = [2.0, 3.0, 4.0, 5.0]$ TeV, while the masses of the W bosons are fixed to their SM value of $m_W = 80$ GeV.
In conclusion, the third process studied is the decay of a heavy W' boson into a W and a Z BSM bosons, with single masses at $m_{W'} = 3.0$ TeV, $m_W = 200$ GeV and $m_Z = 400$ GeV.

In all the three cases, bosons are assumed to decay into quarks, forming jets in the final state.
The complete list of signal processes considered in the analysis, together with their corresponding masses and theoretical cross-sections assumed, is reported in Tab. 5.1. The theoretical cross-sections are derived from the ATLAS internal database for $MC23d$ campaign.

Signal Name	Process	Masses [GeV]	Theoretical cross-section [pb]
VcXH	$V \to XH$	$m_V = 2300$, $m_X = 100$	$4.7435 \cdot 10^{-1}$
VcXH	$V \to XH$	$m_V = 3400$, $m_X = 200$	$4.1550 \cdot 10^{-2}$
VcXH	$V \to XH$	$m_V = 5000$, $m_X = 2500$	$1.6359 \cdot 10^{-3}$
VcXH	$V \to XH$	$m_V = 6000$, $m_X = 400$	$2.1811 \cdot 10^{-4}$
VzWW	$V \to WW$	$m_V = 2000$	$3.8476 \cdot 10^{-3}$
VzWW	$V \to WW$	$m_V = 3000$	$3.5969 \cdot 10^{-4}$
VzWW	$V \to WW$	$m_V = 4000$	$4.5256 \cdot 10^{-5}$
VzWW	$V \to WW$	$m_V = 5000$	$6.2648 \cdot 10^{-6}$
Wprime	$W' \to WZ$	$m_{W'} = 3000$, $m_W = 200$, $m_Z = 400$	$9.6982 \cdot 10^{-16}$
DarkJet	$Z' \to q_D \bar{q}_D$	$m_{Z'} = 3500$, $m_{q_D} = 20$, $m_{q_D} = 20$	$1.05853 \cdot 10^{-9}$

Table 5.1 Signal processes considered in the analysis, with their corresponding masses and theoretical cross-sections.

5.1.3 Background processes

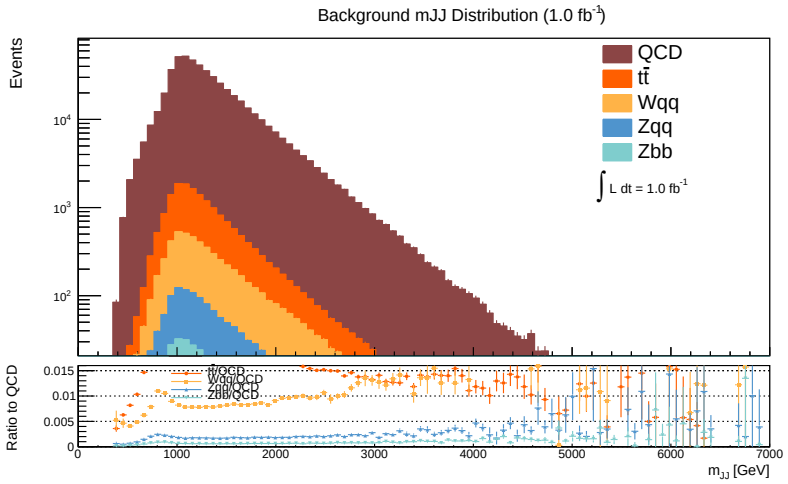


Figure 5.2 Stacked distribution of the di-jet invariant mass m_{JJ} for the dominant background processes considered in the analysis.

5.2 Machine Learning

5.3 Graph Neural Networks

Particle physics events inherently feature variable numbers of reconstructed objects—such as tracks, topological clusters, or jet constituents—interacting within complex geometrical spaces. Graph Neural Networks (GNNs) exploit this structure by representing collision events as sparse directed graphs $G = (V, E)$, where nodes $v_i \in V$ correspond to individual particles or sub-detectors, and edges $e_{ij} \in E$ parameterize relational features or spatial separations $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$.

Node feature vectors $h_i^{(0)}$ encode basic spatial and kinematic observables, including transverse momentum p_T, pseudorapidity η, azimuthal angle ϕ, and impact parameters. Graph neural architectures process relational topology through Message Passing Neural Network (MPNN) frameworks. At execution step t, node representations are updated iteratively using localized neighborhood information:

$$m_i^{(t+1)} = \sum_{j \in \mathcal{N}(i)} \mathcal{M}_\theta \left(h_i^{(t)}, h_j^{(t)}, e_{ij} \right) \tag{5.1}$$

$$h_i^{(t+1)} = \mathcal{U}_\phi \left(h_i^{(t)}, m_i^{(t+1)} \right) \tag{5.2}$$

where $\mathcal{N}(i)$ defines the set of immediate spatial neighbors of node i, $\mathcal{M}_\theta$ represents a parameterized message function, and $\mathcal{U}_\phi$ constitutes a node-state update function.

Permutation invariance across input node orderings is enforced via global pooling operations:

$$h_G = \bigoplus_{v_i \in V} h_i^{(T)} \tag{5.3}$$

where $\bigoplus$ represents a symmetric aggregation operator such as a element-wise summation, mean, or attention-weighted sum. In the context of large-radius jet substructure analyses within ATLAS, GNN architectures capture complex inter-particle correlations, color-flow dynamics, and energy distributions without relying on manually engineered substructure variables such as N-subjettiness τ_N or energy correlation functions D_2.

5.4 Anomaly Detection Implementation with Deep Learning

Deep-learning-based Anomaly Detection in High Energy Physics is operationalized through distinct architectural formulations:

Self-Supervised Reconstruction Architectures: Autoencoders (AEs) compress input vectors $x \in \mathbb{R}^d$ into a lower-dimensional latent representation $z \in \mathbb{R}^k$ ($k \ll d$) via an encoding network $f_\theta(x)$, subsequently reconstructing the input via a decoding network $g_\phi(z)$. Trained exclusively on dominant SM background events using loss metrics such as Mean Squared Error:

$$\mathcal{L}_{\text{MSE}} = \|x - g_\phi(f_\theta(x))\|^2 \tag{5.4}$$

the AE optimizes loss metrics for standard topologies. When evaluated on BSM topologies absent from training ensembles, the AE fails to accurately reconstruct the anomalous features, yielding significantly elevated reconstruction errors. These errors directly construct a continuous Anomaly Score (AS). Variational Autoencoders (VAEs) regularize this latent space by enforcing a standard Gaussian prior distribution $p(z) \sim \mathcal{N}(0, I)$ through a Kullback-Leibler divergence term $\mathcal{D}_{\mathrm{KL}}$, balancing latent space continuity against reconstruction fidelity.

Weakly-Supervised Overdensity Estimation: The Classification Without Labels (CWoLa) paradigm isolates overdensities in resonant spectra by establishing mixed signal-plus-background datasets. Defining a Signal Region (SR) in a invariant mass continuum m_{JJ} alongside adjacent Control Sidebands (SB), a classifier is trained directly on unlabeled experimental data to distinguish SR events from SB events. By the Neyman-Pearson Lemma, a classifier trained on mixed ensembles with differing signal proportions converges monotonically to the optimal Likelihood Ratio:

$$R(x) = \frac{p_{\mathrm{data}}(x|m \in \mathrm{SR})}{p_{\mathrm{bg}}(x|m \in \mathrm{SR})} \tag{5.5}$$

Advanced weakly-supervised density estimators, such as CATHODE, CURTAINS, and ANODE, employ conditional normalizing flows or optimal transport algorithms to model sideband background dynamics, interpolating background probability densities directly into the SR to eliminate feature correlations and artificial mass sculpting.

In the ATLAS ADJJ (Anomaly Detection in Dijet final states) analysis, GNN architectures are interfaced directly with model-independent search frameworks. Unsupervised GNN autoencoders process the constituent point-clouds of large-radius anti-k_t ($R = 1.0$) jets, outputting an Anomaly Score for every event. Selecting thresholds on the AS distribution defines specific Working Points (WPs), achieving background rejection factors spanning orders of magnitude.

Model-independent signal searches are subsequently executed by scanning the di-jet invariant mass m_{JJ} spectrum across defined AS WPs. Smooth parametric functions model the suppressed SM background background continuum, while potential anomalous signals are evaluated using flexible resonance fits, such as Double-Sided Crystal Ball functions. Binned maximum likelihood fits combined with the CL_s statistical method derive 95% Confidence Level upper limits on production cross-sections, providing a sensitivity bridge between deep graph representations and model-agnostic BSM discoveries at CERN.

5.5 Overview of Anomaly Detection

Anomaly Detection encompasses algorithms that identify data instances deviating significantly from nominal distributions [**?** **?**]. In standard industrial contexts, anomalies manifest as out-of-sample or off-manifold events, such as fraudulent transactions or equipment failures [**?** **?**]. In High Energy Physics (HEP), single events rarely provide statistical evidence of new phenomena; BSM signals typically manifest as subtle over-densities in phase space [**?** **?**].

Within particle physics, ML-based AD strategies operate across different levels of supervision [? ?]. Weakly supervised methods, such as Classification Without Labels (CWoLa) and CATHODE, estimate localized over-densities by comparing signal-enriched regions against adjacent sidebands [? ?]. Conversely, fully unsupervised methods operate solely on background processes or unlabeled experimental data [? ?]. These algorithms isolate rare topology features in kinematic distributions, providing an unbiased tool for model-agnostic BSM searches [? ?].

5.6 Graph Neural Networks

Standard feedforward architectures struggle to process variable-size physical objects without imposing artificial ordering [? ?]. Hadronic jets, composed of varying counts of collimated sub-particles, naturally map to graph structures $G = (V, E)$, where V denotes constituent nodes and E represents interaction edges [? ?]. Node features contain kinematic variables $\mathbf{x}_i = (p_T, \eta, \phi)$, while edge features capture pairwise angular separations $\Delta R = \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ [? ?].Graph Neural Networks (GNNs) preserve permutation invariance while executing local message passing [? ?]. The representation $\mathbf{h}_i^{(l)}$ of node i at layer l updates according to aggregated messages from its spatial neighborhood $\mathcal{N}(i)$ [?]:

$$\mathbf{m}_{ij}^{(l)} = \phi_e \left(\mathbf{h}_i^{(l-1)}, \mathbf{h}j^{(l-1)}, \mathbf{e}ij \right) \tag{5.6}$$

$$\mathbf{h}i^{(l)} = \phi_v \left(\mathbf{h}i^{(l-1)}, \bigoplus j \in \mathcal{N}(i) \mathbf{m}ij^{(l)} \right) \tag{5.7}$$

where ϕ_e and ϕ_v denote neural operations, $\mathbf{e}_{ij}$ is the edge feature, and $\bigoplus$ represents a permutation-invariant aggregation operator such as sum, mean, or max [? ?]. Graph-level outputs obtain global representations via permutation-invariant pooling functions across all nodes [?].

5.7 Anomaly Detection Implementation via Deep Architectures

Unsupervised AD implementation utilizing DNNs and GNNs relies on latent representation learning [? ?]. Autoencoders (AEs) compress input graph representations $\mathbf{x}$ into a lower-dimensional bottleneck $\mathbf{z} = f_{\text{enc}}(\mathbf{x})$ before generating a reconstructed output $\hat{\mathbf{x}} = f_{\text{dec}}(\mathbf{z})$ [? ?]. When trained exclusively on dominant Standard Model background events, the network learns lossy compression tailored to nominal physics patterns [? ?].Events containing BSM physics produce higher reconstruction errors due to unlearned substructure features [? ?]. An Anomaly Score (AS) is defined using the Mean Squared Error (MSE) metric [? ?]:

$$AS = \frac{1}{N} \sum_{i=1}^{N} |\mathbf{x}_i - \hat{\mathbf{x}}_i|^2 \tag{5.8}$$

In ATLAS model-independent searches, cuts applied to the calculated AS distribution isolate BSM candidates [? ?]. By filtering out high-probability background interactions, these architectures define signal-enriched regions directly from data, enabling model-agnostic resonance searches [? ?].

5.8 CURCH

5.9 How is it implemented?

The process of training a DL model is divided into three main phases: training, validation and test loop. They are defined "loop" because in each of them a different section of the dataset is used and the code loops over it.
The most important stage is the training loop: in our case the chosen DL model is trained to reconstruct arrays containing information about particle jets, using only data with normal behaviour (background events) [57]. In this phase, parameters of the model are adjusted in order to best fit to data. By doing so, DL model is able to recognise very well normal data and their structure. Our hope is that, once it faces an anomaly (signal event), it is able to identify it due to its difference from what it knows. The second step is the validation loop, where a valuation of the performance of the model is elaborated using the validation set. The valuation and the search for the best weights of the model are executed computing distance between output and input over training and validation set. When this distance is minimum within the validation loop, we save the model's weights.
The trained and tuned model is then tested, using a dataset containing unseen data, the test set, where there are both background and signal events. The purpose of this phase is to validate results previously obtained and get a preview of what will happen once the model will be running with completely new data. The task is completed computing the Anomaly Score (AS). In this thesis, we implement it as a distance between input and output, with different dimension: instead of calculating it over a batch of data, we do so for each event. In this way we are able to plot a reconstruction probability graph: we expect that the distributions of AS from background and signal events are different, because signal data contains anomalies which our model does not know. By means of these distributions we can identify a threshold of separation between them. For any event we pass to our trained model, we compute AS: if it higher (lower) than the chosen threshold, we are looking at a background (signal) event. Since AD is distinguishing signal events from background, we are basically executing a classification task with two classes. Implementation of Anomaly Detection is discussed more in depth in chapters **??** and **??**.

Chapter 6

Conclusions

Bibliography

[1] ATLAS Collaboration. *"Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC"*. *Phys. Lett. B*, 716:1–29, (2012).

[2] CMS Collaboration. *"Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC"*. *Phys. Lett. B*, 716:30–61, (2012).

[3] K. Garrett et al. *"Dark Matter: A Primer"*. *Advances in Astronomy*, (2011). doi: 10.1155/2011/968283.

[4] M. E. Peskin and D. V. Schroeder. *"An Introduction to quantum field theory"*. (1995). doi: 10.1201/9780429503559.

[5] Wikipedia The Free Encyclopedia. *"Standard Model"*, (2026). URL `https://en.wikipedia.org/w/index.php?title=Standard_Model&oldid=1368397978`.

[6] R. P. Feynman. *"A New Approach to Quantum Theory"*. Ed. by Brown Laurie M. World Scientific Publishing Co. Pte. Ltd., (2005).

[7] A. Pich. *"The Standard Model of Electroweak Interactions"*, (2012). URL `https://arxiv.org/abs/1201.0537`.

[8] E. Noether. *"Invariant variation problems"*. *Transport Theory and Statistical Physics*, (1971). URL `http://dx.doi.org/10.1080/00411457108231446`.

[9] W. Greiner et al. *"Quantum Chromodynamics"*. (2002). URL `https://doi.org/10.1007/978-3-662-04707-1`.

[10] S. Weinberg. *"A Model of Leptons"*. (1967). URL `https://link.aps.org/doi/10.1103/PhysRevLett.19.1264`.

[11] F. Takahashi et al. Particle Data Group. *"Review of Particle Physics"*. Table 11, (2026). URL `https://pdg.lbl.gov/2026/web/viewer.html?file=../reviews/rpp2026-rev-higgs-boson.pdf`.

[12] M. K. Gaillard J. Ellis and D. V. Nanopoulos. *"A Historical Profile of the Higgs Boson"*. (2016). URL `https://cds.cern.ch/record/2012465`.

[13] ATLAS Collaboration. *"Search for heavy neutral Higgs bosons decaying into a top quark pair in 140 fb^{-1} of proton-proton collision data at $\sqrt{s}$= 13 TeV with the ATLAS detector"*. *Journal of High Energy Physics*, 13, (2024).

[14] A. Strumia M. Cirelli and J. Zupan. *"Dark Matter"*, (2025). URL https://arxiv.org/abs/2406.01705.

[15] E. Hubble and M. L. Humason. *"The Velocity-Distance Relation among Extra-Galactic Nebulae"*. (1931). URL https://ui.adsabs.harvard.edu/abs/1931ApJ....74...43H.

[16] R. D. Blandford T. G. Brainerd and I. Smail. *"Weak Gravitational Lensing by Galaxies"*. (1996).

[17] D. Clowe et al. *"A Direct Empirical Proof of the Existence of Dark Matter"*. (2006). URL http://dx.doi.org/10.1086/508162.

[18] D. Alves et al. *"Simplified Models for LHC New Physics Searches"*. *J. Phys. G*, (2012).

[19] R. Torre D. Pappadopulo, A. Thamm and A. Wulzer. *"Heavy vector triplets: bridging theory and data"*. *Journal of High Energy Physics*, (2014).

[20] G. et al. Aad. *"Anomaly detection search for new resonances decaying into a Higgs boson and a generic new particle X in hadronic final states using* $\sqrt{s} = 13$ TeV *pp collisions with the ATLAS detector"*. *Phys. Rev. D*, (2023). URL https://link.aps.org/doi/10.1103/PhysRevD.108.052009.

[21] A. Tumasyan et al. *"Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at s=13TeV"*. *Physics Letters, Section B: Nuclear, Elementary Particle and High-Energy Physics*, (2023). doi: 10.1016/j.physletb.2023.137813.

[22] G. Albouy et al. *"Theory, phenomenology, and experimental avenues for dark showers: a Snowmass 2021 report"*. *The European Physical Journal C*, (12). URL http://dx.doi.org/10.1140/epjc/s10052-022-11048-8.

[23] G. Aad et al. *"Search for resonant production of dark quarks in the dijet final state with the ATLAS detector"*. *Journal of High Energy Physics*, (2024). URL https://doi.org/10.1007/JHEP02(2024)128.

[24] E. Mobs. *"The CERN accelerator complex in 2019"*. (2019). URL https://cds.cern.ch/record/2684277.

[25] Communications Education and CERN Outreach Group. *"LHC the guide FAQ"*. URL https://cds.cern.ch/record/2255762.

[26] L. Evans and P. Bryant. *"LHC Machine"*. *Journal of Instrumentation*, (2008). URL https://doi.org/10.1088/1748-0221/3/08/S08001.

[27] L. Rossi. *"The LHC superconducting magnets"*. *Conf. Proc. C*, (2003).

[28] O. Brüning et al. *"LHC Design Report"*. CERN Yellow Reports: Monographs. (2004). URL https://cds.cern.ch/record/782076.

[29] ATLAS Collaboration. *"Cumulative integrated luminosity delivered to and recorded by the ATLAS Experiment during LHC Run 1, 2 and 3"*. (2026). URL https://cds.cern.ch/record/2964765.

[30] T. M. Yan and S. D. Drell. *"The parton model and its applications"*. *International Journal of Modern Physics A*, (2014). URL http://dx.doi.org/10.1142/S0217751X14300713.

[31] Jennet Elizabeth Dickinson. *"ATLAS Measurements of the Higgs Boson Coupling to the Top Quark in the Higgs to Diphoton Decay Channel"*, pages 5–27. Springer International Publishing, (2021). URL https://doi.org/10.1007/978-3-030-86368-5_2.

[32] Tobias Bisanz. *"Operational experience and performance with the ATLAS pixel detector at the Large Hadron Collider at CERN"*. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 1091, (2026). URL https://www.sciencedirect.com/science/article/pii/S0168900226004869.

[33] ATLAS collaboration. The ATLAS Experiment. URL: https://atlas.cern/.

[34] Riccardo Maria Bianchi and ATLAS Collaboration. *"ATLAS experiment schematic or layout illustration"*. (2022). URL https://cds.cern.ch/record/2837191.

[35] *"The ATLAS Experiment at the CERN Large Hadron Collider"*. *Journal of Instrumentation*, (2008). URL https://doi.org/10.1088/1748-0221/3/08/S08003.

[36] R.L. Gluckstern. *"Uncertainties in track momentum and direction, due to multiple scattering and measurement errors"*. *Nuclear Instruments and Methods*, 24: 381–389, (1963). URL https://www.sciencedirect.com/science/article/pii/0029554X63903471.

[37] *"ATLAS inner detector"*. Technical design report. ATLAS. CERN, Geneva, (1997). URL https://cds.cern.ch/record/331063.

[38] ATLAS Collaboration. *"Experiment Briefing: Keeping the ATLAS Inner Detector in perfect alignment"*. (2020). URL https://cds.cern.ch/record/2723878.

[39] *"ATLAS calorimeter performance"*. Technical design report. ATLAS. CERN, (1996). URL https://cds.cern.ch/record/331059.

[40] *"ATLAS liquid-argon calorimeter"*. Technical design report. ATLAS. CERN, (1996). URL https://cds.cern.ch/record/331061.

[41] G. Aad et al. *"The ATLAS experiment at the CERN Large Hadron Collider: a description of the detector configuration for Run 3"*. *Journal of Instrumentation*, 19(05), (2024). URL https://doi.org/10.1088/1748-0221/19/05/P05063.

[42] G. Aad et al. *"Technical Design Report for the Phase-I Upgrade of the ATLAS TDAQ System"*. Technical report, (2013). URL https://cds.cern.ch/record/1602235.

[43] *"ATLAS Computing"*. Technical design report. ATLAS. CERN, Geneva, (2005). URL https://cds.cern.ch/record/837738.

[44] S. Agostinelli et al. *"Geant4—a simulation toolkit"*. *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment*, 506(3):250–303, (2003). URL https://www.sciencedirect.com/science/article/pii/S0168900203013688.

[45] T. G. Cornelissen et al. *"Updates of the ATLAS Tracking Event Data Model (Release 13)"*. Technical report, CERN, (2007). URL https://cds.cern.ch/record/1038095.

[46] G. Aad et al. *"Topological cell clustering in the ATLAS calorimeters and its performance in LHC Run 1"*. *The European Physical Journal C*, (2017). URL http://dx.doi.org/10.1140/epjc/s10052-017-5004-5.

[47] M. Aaboud et al. *"Jet reconstruction and performance using particle flow with the ATLAS Detector"*. *The European Physical Journal C*, (2017). URL https://doi.org/10.1140/epjc/s10052-017-5031-2.

[48] ATLAS collaboration. *"Improving jet substructure performance in ATLAS using Track-CaloClusters"*. (2017).

[49] ATLAS collaboration. *"R=0.4 jets input comparison and Monte Carlo calibration with the ATLAS Detector"*. Technical report, CERN, (2022). URL https://cds.cern.ch/record/2824558.

[50] M. Cacciari, G. P. Salam, and G. Soyez. *"The anti-ktjet clustering algorithm"*. *Journal of High Energy Physics*, (2008). URL http://dx.doi.org/10.1088/1126-6708/2008/04/063.

[51] J. Thaler and K. Van Tilburg. *"Identifying boosted objects with N-subjettiness"*. *Journal of High Energy Physics*, (2011). URL http://dx.doi.org/10.1007/JHEP03(2011)015.

[52] A. J. Larkoski, G. P. Salam, and J. Thaler. *"Energy correlation functions for jet substructure"*. *Journal of High Energy Physics*, (2013). URL https://doi.org/10.1007/JHEP06(2013)108.

[53] A. J. Larkoski, I. Moult, and D. Neill. *"Analytic Boosted Boson Discrimination"*. (2016). URL https://arxiv.org/abs/1507.03018.

[54] G. Kasieczka et al. *"The LHC Olympics 2020 a community challenge for anomaly detection in high energy physics"*. *Reports on Progress in Physics*, (2021). URL http://dx.doi.org/10.1088/1361-6633/ac36b9.

[55] E. Gross and O. Vitells. *"Trial factors for the look elsewhere effect in high energy physics"*. *The European Physical Journal C*, (2010). URL http://dx.doi.org/10.1140/epjc/s10052-010-1470-8.

[56] V. Chandola, A. Banerjee, and V. Kumar. *"Anomaly detection: A survey"*. *ACM Computing Surveys*, 41, (2009). doi: 10.1145/1541880.1541882.

[57] Cloudera. Deep Learning for Anomaly Detection. URL: https://ff12.fastforwardlabs.com/.

[58] F. Nuzzo. *"Sanity Checks for Explanations of Deep Neural Networks Predictions"*, (2020). URL https://webthesis.biblio.polito.it/15595/.

[59] S. Han. *"Efficient Methods and Hardware for Deep Learning"*. PhD thesis, Stanford University, Stanford, CA, USA, (2017).

[60] T.S. Roy et A.H. Vijay. A robust anomaly finder based on autoencoders, 2020. URL https://arxiv.org/abs/1903.02032.

[61] H. Kaiming, Z. Xiangyu, R. Shaoqing, and S. Jian. *"Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification"*, (2015). URL https://arxiv.org/abs/1502.01852.

[62] Scikit-learn. AUC. URL: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.auc.html, .

[63] Scikit-learn. ROC curve. URL: https://scikit-learn.org/stable/modules/generated/sklearn.metrics.roc_curve.html, .

[64] V. Belis, P. Odagiu, and Thea K. Aarrestad. *"Machine learning for anomaly detection in particle physics"*. *Reviews in Physics*, (2023). URL http://dx.doi.org/10.1016/j.revip.2024.100091.